Swimming Filaments!!

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1sectionsection=0 . Micropolar media

In this section we consider micropolar media whose microstates take values in the group of rigid transformations, which we choose to represent as

$$SE(3) = \left\{ \begin{bmatrix} Q & r \\ 0 & 1 \end{bmatrix} \in GL(4) \middle| Q \in SO(3), r \in \mathbb{R}^3 \right\}. \tag{1}$$

The state of such a medium can be decomposed Components of tensors on $SE(3)$ will have Greek indices,

$$T^{\alpha}_{\beta\gamma}, \chi^{\mu}_{\nu}, \text{ etc}\ldots, \tag{2}$$

which take values in $\mathbb{Z}_6 = \{0,1,2,3,4,5\}$,

$$T^{\alpha}_{\beta\gamma}\chi^{\beta}_{\alpha} := \sum_{\beta,\alpha=0}^{5} T^{\alpha}_{\beta\gamma}\chi^{\beta}_{\alpha}. \tag{3}$$

Spacetime coordinates $s = (s^0 = t, s^1, \ldots, s^p)$, consisting of a single timelike coordinate $s^0 = t$ and p spacelike material coordinates $(s^1, \ldots, s^p)$ are used throughout. Components of tensors on spacetime will have latin indices,

$$F^{a}_{bc} e^{m}_{n}, \text{ etc}\ldots, \tag{4}$$

which take values in $\mathbb{Z}_{p+1} = \{0, \ldots, p\}$

$$F^{a}_{bc} e^{b}_{a} := \sum_{a,b=0}^{p} F^{a}_{bc} e^{b}_{a}. \tag{5}$$

The state of a micropolar medium on the spacetime manifold $S \subset \mathbb{R}^{p+1}$ is a map

$$\rho : S \to SE(3) \tag{6}$$

which assigns a microstate to each point in S. Kinematical constraints due to the geometry of $SE(3)$ are discussed in 0.1. We introduce a sigma-model action in 0.2 and outline conditions for extremality. Explicit equations of motion for the Kirchoff rod, an example of a $p = 1$ micropolar material, are found in 0.3.

1.1subsectionsection=0 . Kinematics and geometry

1.1.1subsubsectionsection=0 . Spacetime

Let $S \subset \mathbb{R}^{p+1}$ be a compact and simply connected domain with coordinates

$$(s^0, s^1, \ldots, s^p) = (t, s^1, \ldots, s^p) \tag{7}$$

and Minkowski metric with signature $(+, -, \ldots, -)$

$$\left\langle \mathrm{d}s^i \middle| \mathrm{d}s^j \right\rangle = \begin{cases} -1 & \text{if } i = j > 0 \\ 1 & \text{if } i = j = 0 \\ 0 & \text{if } i \neq j \end{cases} \tag{8}$$

Choose your favorite matrix representations of the Lie algebras $\mathbb{R}^3$, $\mathfrak{so}(3)$, and $\mathfrak{se}(3)$:

1.1.2subsubsectionsection=0 . $\mathbb{R}^3$

- Basis:

$$e_0, e_1, e_2 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \tag{9}$$

- Inner product:

$$\langle a|b\rangle_{\mathbb{R}^3} := \mathrm{tr}\left(a^T b\right) = a^T b \tag{10}$$

- Metric components:

$$\delta_{\alpha\beta} := \left\langle e_\alpha \middle| e_\beta \right\rangle_{\mathbb{R}^3} \tag{11}$$

- Lie bracket:

$$[a, b]_{\mathbb{R}^3} := a \times b \text{ (cross product)} \tag{12}$$

- Structure constants:

$$\varepsilon_{\alpha\beta\gamma} := \left\langle e_\alpha \middle| [e_\beta, e_\gamma] \right\rangle \tag{13}$$

1.1.3subsubsectionsection=0 . $\mathfrak{so}(3)$

- Basis

$$\hat{e}_0, \hat{e}_1, \hat{e}_2 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{bmatrix}, \begin{bmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \tag{14}$$

- Inner product

$$\langle \hat{a} | \hat{b} \rangle_{\mathfrak{so}(3)} := \mathrm{tr}\left(\hat{a}^T \hat{b}\right) \tag{15}$$

- Metric components

$$\langle \hat{e}_\alpha | \hat{e}_\beta \rangle = 2\delta_{\alpha\beta} \tag{16}$$

- Lie bracket

$$\left[\hat{a}, \hat{b}\right]_{\mathfrak{so}(3)} := \hat{a}\hat{b} - \hat{b}\hat{a} \tag{17}$$

- Structure constants

$$\langle \hat{e}_\alpha | [\hat{e}_\beta, \hat{e}_\gamma] \rangle = 2\varepsilon_{\alpha\beta\gamma} \tag{18}$$

- Action on $\mathbb{R}^3$

$$\langle e_\alpha | \hat{e}_\beta | e_\gamma \rangle = \varepsilon_{\alpha\beta\gamma} \tag{19}$$

1.1.4subsubsectionsection=0 . $\mathfrak{se}(3)$

- Basis

$$\phi_0, \phi_2, \phi_4 = \begin{bmatrix} \hat{e}_0 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \hat{e}_1 & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} \hat{e}_2 & 0 \\ 0 & 0 \end{bmatrix} \tag{20}$$

$$\phi_1, \phi_3, \phi_5 = \begin{bmatrix} 0 & e_0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & e_1 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & e_2 \\ 0 & 0 \end{bmatrix} \tag{21}$$

- Inner product

$$\langle a|b\rangle_{\mathfrak{se}(3)} := \mathrm{tr}\left(a^T b\right) \tag{22}$$

- Covariant metric components

$$g_{\alpha\beta} := \left\langle \phi_\alpha \middle| \phi_\beta \right\rangle \tag{23}$$

$$g_{\alpha\beta} = \frac{3+(-1)^\alpha}{2}\delta_{\alpha\beta} = \begin{cases} 2\delta_{\alpha\beta} & \text{if } \alpha,\beta \in \{0,2,4\} \\ \delta_{\alpha\beta} & \text{if } \alpha,\beta \in \{1,3,5\} \\ 0 & \text{otherwise} \end{cases} \tag{24}$$

- Contravariant metric components

$$\left[g^{\alpha\beta}\right] := \left[g_{\alpha\beta}\right]^{-1} \text{ (matrix inverse)} \tag{25}$$

$$g_{\alpha\beta} = \frac{2}{3+(-1)^\alpha}\delta_{\alpha\beta} = \begin{cases} \frac{1}{2}\delta_{\alpha\beta} & \text{if } \alpha,\beta \in \{0,2,4\} \\ \delta_{\alpha\beta} & \text{if } \alpha,\beta \in \{1,3,5\} \\ 0 & \text{otherwise} \end{cases} \tag{26}$$

- Lie bracket

$$[\mu,\nu]_{\mathfrak{se}(3)} := \mu\nu - \nu\mu \tag{27}$$

- Structure constants

$$\begin{aligned} c_{\alpha\beta\gamma} &:= \langle \phi_\alpha | [\phi_\beta, \phi_\gamma] \rangle \\ c_{2\alpha,2\beta,2\gamma} &= 2\varepsilon_{\alpha\beta\gamma} \\ c_{2\alpha+1,2\beta,2\gamma+1} &= c_{2\alpha+1,2\beta+1,2\gamma} = \varepsilon_{\alpha\beta\gamma} \\ c_{2\alpha,2\beta,2\gamma+1} &= c_{2\alpha,2\beta+1,2\gamma} = c_{2\alpha,2\beta+1,2\gamma+1} = c_{2\alpha+1,2\beta,2\gamma} = c_{2\alpha+1,2\beta+1,2\gamma+1} = 0 \end{aligned} \tag{28}$$

Where $\alpha,\beta,\gamma \in \{0,1,2\}$ and $\varepsilon_{\alpha\beta\gamma}$ is the permutation symbol on $\{0,1,2\}$.

- Adjoint action

$$\hat{\mu}\,|\nu\rangle := |[\mu,\nu]\rangle \text{ for } \mu,\nu \in \mathfrak{se}(3) \tag{29}$$

$$\langle \phi_\alpha | \hat{\phi}_\beta | \phi_\gamma \rangle = c_{\alpha\beta\gamma} \tag{30}$$

$$\tag{31}$$

$$\langle \phi^\alpha | \hat{\phi}_\beta | \phi_\gamma \rangle = c^\alpha_{\beta\gamma} \tag{32}$$

$$\tag{33}$$

- Some useful formulas

$$\phi_{2\alpha}, \phi_{2\alpha+1} = \begin{bmatrix} \hat{e}_\alpha & 0 \\ 0 & 0 \end{bmatrix}, \begin{bmatrix} 0 & e_\alpha \\ 0 & 0 \end{bmatrix} \tag{34}$$

$$\phi_{2\beta} = \sum_{\alpha,\gamma\in\mathbb{Z}_3} \varepsilon_{\alpha\beta\gamma}\phi_{2\alpha+1}\phi^T_{2\gamma+1} \text{ for } \beta \in \mathbb{Z}_3 \tag{35}$$

$$\phi^0, \phi^2, \phi^4 = \frac{1}{2}\phi_0, \frac{1}{2}\phi_2, \frac{1}{2}\phi_4 \tag{36}$$

$$\phi^1, \phi^3, \phi^5 = \phi_1, \phi_3, \phi_5 \tag{37}$$

$$\hat{\phi}_{2\alpha}\phi_{2\beta} = \varepsilon_{\alpha\beta\gamma}\phi_{2\gamma} \tag{38}$$

$$\hat{\phi}_{2\alpha}\phi_{2\beta+1} = \varepsilon_{\alpha\beta\gamma}\phi_{2\gamma+1} \tag{39}$$

$$\hat{\phi}_{2\alpha+1}\phi_{2\beta} = \varepsilon_{\alpha\beta\gamma}\phi_{2\gamma+1} \tag{40}$$

$$\hat{\phi}_{2\alpha+1}\phi_{2\beta+1} = 0 \tag{41}$$

$$c^{2\alpha}_{2\beta,2\gamma} = c^{2\alpha+1}_{2\beta,2\gamma+1} = c^{2\alpha+1}_{2\beta+1,2\gamma} = \varepsilon_{\alpha\beta\gamma} \tag{42}$$

$$c^{2\alpha+1}_{2\beta+1,2\gamma+1} = 0 \tag{43}$$

$$\left(c^{\dagger}\right)^{\alpha}_{\beta\gamma} = \langle \phi^{\alpha} | \hat{\phi}^{\dagger}_{\beta} | \phi_{\gamma} \rangle \tag{44}$$

$$= \langle \hat{\phi}_{\beta} \phi^{\alpha} | \phi_{\gamma} \rangle \tag{45}$$

$$= \langle \phi_{\gamma} | \hat{\phi}_{\beta} \phi^{\alpha} \rangle \tag{46}$$

$$= g^{\alpha\mu} g_{\gamma\nu} \langle \phi^{\nu} | \hat{\phi}_{\beta} | \phi_{\mu} \rangle \tag{47}$$

$$= \sum_{\mu,\nu=0}^{5} \frac{2}{3+(-1)^{\alpha}} \delta_{\alpha\mu} \frac{3+(-1)^{\gamma}}{2} \delta_{\gamma\nu} c^{\nu}_{\beta\mu} \tag{48}$$

$$= \sum_{\mu,\nu=0}^{5} \frac{3+(-1)^{\gamma}}{3+(-1)^{\alpha}} \delta_{\alpha\mu} \delta_{\gamma\nu} c^{\nu}_{\beta\mu} \tag{49}$$

$$= \frac{3+(-1)^{\gamma}}{3+(-1)^{\alpha}} c^{\gamma}_{\beta\alpha} \text{ (no sum over repeated indices)} \tag{50}$$

$$\frac{3+(-1)^{\gamma}}{3+(-1)^{\alpha}} = \begin{cases} 1 & \text{if } \alpha+\gamma \text{ is even} \\ 2 & \text{if } \alpha \text{ is odd }, \gamma \text{ is even} \\ 1/2 & \text{if } \gamma \text{ is odd }, \alpha \text{ is even} \end{cases} \tag{51}$$

1.1.5subsubsectionsection=0 . Matrix-valued forms

The basis (20-21) and inner product (22) induce a left invariant frame field and a left invariant metric on $SE(3)$

$$T_\rho SE(3) = \operatorname{span}\{\rho\phi_\alpha\}_{\alpha=0}^{5},\ \forall \rho \in SE(3) \tag{52}$$

$$\left\langle \rho\phi_\alpha \middle| \rho\phi_\beta \right\rangle := \left\langle \phi_\alpha \middle| \phi_\beta \right\rangle_{\mathfrak{se}(3)} \tag{53}$$

Define $\mathscr{G}$ to be the $SE(3)$ bundle over S and define $\mathfrak{g}^*$ to be the $\mathfrak{se}(3)$ bundle over T^*S. Any smooth section of $\mathscr{G}$ is determined up to a constant $SE(3)$ matrix, by a unique section section of $\mathfrak{g}^*$,

$$\rho : S \to SE(3) \tag{54}$$

$$\psi : S \to \mathfrak{se}(3) \otimes T^*S \tag{55}$$

$$\psi = \rho^{-1}\,\mathrm{d}\rho \tag{56}$$

$$= \psi_i\,\mathrm{d}s^i \tag{57}$$

$$= \psi_i^\alpha \phi_\alpha\,\mathrm{d}s^i\,. \tag{58}$$

which satisfies the structure equation

$$0 = \mathrm{d}^2\rho = \rho(\mathrm{d}\psi + \psi \wedge \psi) \tag{59}$$

Or, in local coordinates

$$\partial_i \left|\psi_j\right\rangle - \partial_j \left|\psi_i\right\rangle + \hat{\psi}_i \left|\psi_j\right\rangle = 0 \tag{60}$$

We can introduce a metric $\langle\cdot|\cdot\rangle$ on $\mathfrak{g}^*$ by taking the tensor product of the left invariant metric on $SE(3)$ with the minkowski metric on S.

$$\left\langle \rho\phi_\alpha \,\mathrm{d}s^i \middle| \rho\phi_\alpha \,\mathrm{d}s^i \right\rangle := \left\langle \phi_\alpha \middle| \phi_\beta \right\rangle_{\mathfrak{se}(3)} \left\langle \mathrm{d}s^i \middle| \mathrm{d}s^j \right\rangle_S \tag{61}$$

The bundle $\mathfrak{g}^*$, decomposes with respect to (61), into an orthogonal direct sum of space-like angular, time-like angular, space-like linear, and time-like linear subspaces:

$$\left|\psi\right\rangle = \sum_{\alpha \text{ even}} \sum_{k=1}^{p} \psi^\alpha_k \left|\phi_\alpha \,\mathrm{d}s^k\right\rangle \qquad \text{(space-like angular)} \tag{62}$$

$$+ \sum_{\alpha \text{ even}} \psi^\alpha_0 \left|\phi_\alpha \,\mathrm{d}s^0\right\rangle \qquad \text{(time-like angular)} \tag{63}$$

$$+ \sum_{\alpha \text{ odd}} \sum_{k=1}^{p} \psi^\alpha_k \left|\phi_\alpha \,\mathrm{d}s^k\right\rangle \qquad \text{(space-like linear)} \tag{64}$$

$$+ \sum_{\alpha \text{ odd}} \psi^\alpha_0 \left|\phi_\alpha \,\mathrm{d}s^0\right\rangle \qquad \text{(time-like linear)} \tag{65}$$

1.2subsectionsection=0 . Dynamics

1.2.1subsubsectionsection=0 . $\mathfrak{se}(3)$ sigma-model action

Let E be a functional of the form

$$E[\rho] = \int_S \frac{1}{2} \left\langle \mathrm{d}\rho \middle| B(\rho, s) \middle| \mathrm{d}\rho \right\rangle \mathrm{d}^{p+1} s \tag{66}$$

where the tensor

$$B : T^* SE(3) \to T^* SE(3) \tag{67}$$

$$B(\rho) : T^*_\rho SE(3) \to T^*_\rho SE(3) \tag{68}$$

$$B(\rho) = \sum_{\alpha,\beta=0}^{5} \sum_{i,j=0}^{p} B_{ij}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^i\right\rangle \left\langle \rho\phi_\beta \, \mathrm{d}s^j\right| \tag{69}$$

describes inertia and stiffness of the medium. We require that B satisfy the following:

- B is self-adjoint with respect to the metric (61)

$$B = B^\dagger \tag{70}$$

- B is $SE(3)$-equivariant

$$\left\langle \gamma\mu \right| B(\gamma) \left| \gamma\nu \right\rangle = \left\langle \mu \right| B(1) \left| \nu \right\rangle \text{ for all } \gamma \in SE(3) \text{ and } \mu, \nu \in \mathfrak{se}(3) \tag{71}$$

- B preserves the four subspaces mentioned in (62-65), is positive definite when restricted to a timelike subspace, and negative definite when restricted to a spacelike subspace.

$$B = \sum_{\alpha,\beta \text{ even}} \sum_{i,j=1}^{p} B_{ij}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^i\right\rangle \left\langle \rho\phi_\beta \, \mathrm{d}s^j\right| \qquad \text{(Rotational stiffness)} \tag{72}$$

$$+ \sum_{\alpha,\beta \text{ even}} B_{00}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^0\right\rangle \left\langle \rho\phi_\beta \, \mathrm{d}s^0\right| \qquad \text{(Rotational intertia)} \tag{73}$$

$$+ \sum_{\alpha,\beta \text{ odd}} \sum_{i,j=1}^{p} B_{ij}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^i\right\rangle \left\langle \rho\phi_\beta \beta \, \mathrm{d}s^j\right| \qquad \text{(Linear stiffness)} \tag{74}$$

$$+ \sum_{\alpha,\beta \text{ odd}} B_{00}^{\alpha\beta}(s) \left|\rho\phi_\alpha \, \mathrm{d}s^0\right\rangle \left\langle \rho\phi_\beta \beta \, \mathrm{d}s^0\right| \qquad \text{(Mass)} \tag{75}$$

If these conditions are satisfied by B, then the Lagrangian density

$$\mathscr{L} = \frac{1}{2} \left\langle \mathrm{d}\rho \right| B(\rho, s) \left| \mathrm{d}\rho \right\rangle \tag{76}$$

is invariant under the transformation

$$\rho \mapsto \gamma\rho \tag{77}$$

$$\mathrm{d}\rho \mapsto \gamma\,\mathrm{d}\rho \tag{78}$$

for any $\gamma \in SE(3)$, so, without loss of generality we can take $\mathscr{L}$ to be a function of $\psi = \rho^{-1}\,\mathrm{d}\rho$ and s.

$$\mathscr{L}(\rho, \mathrm{d}\rho\,, s) = \mathscr{L}(1, \rho^{-1}\,\mathrm{d}\rho\,, s) = \mathscr{L}(\psi, s) \tag{79}$$

1.2.2subsubsectionsection=0 . Variational derivatives

Suppose $I \subset \mathbb{R}$ is a compact interval, and the fields

$$\rho : S \to SE(3) \tag{80}$$

$$\rho(s) = \begin{bmatrix} Q(s) & r(s) \\ 0 & 1 \end{bmatrix} \tag{81}$$

$$\tilde{\rho} : I \times S \to SE(3) \tag{82}$$

$$\tilde{\rho}(\xi, s) = \begin{bmatrix} \tilde{Q}(\xi, s) & \tilde{r}(\xi, s) \\ 0 & 1 \end{bmatrix} \tag{83}$$

satisfy

$$\tilde{\rho}(0, s) = \rho(s). \tag{84}$$

It follows that the corresponding $\mathfrak{se}(3)$-valued 1-forms

$$\psi = \rho^{-1}\,\mathrm{d}\rho \tag{85}$$

$$\psi(s) = \psi_i(s)\,\mathrm{d}s^i \tag{86}$$

$$\tilde{\psi} = \tilde{\rho}^{-1}\,\mathrm{d}\tilde{\rho} \tag{87}$$

$$\tilde{\psi}(\xi, s) = \tilde{\psi}_i(\xi, s)\,\mathrm{d}s^i + \tilde{\psi}_\xi(\xi, s)\,\mathrm{d}\xi \tag{88}$$

must satisfy

$$\tilde{\psi}(0, s) = \psi(s) + \tilde{\psi}_\xi(0, s)\,\mathrm{d}\xi \tag{89}$$

$$\tilde{\psi}_i(0, s) = \psi_i(s). \tag{90}$$

$$\partial_j \psi_i = \left(\partial_i + \hat{\psi}_i\right)\psi_j \tag{91}$$

$$\partial_\xi \tilde{\psi}_i(0, s) = \left(\partial_i + \hat{\psi}_i(s)\right)\tilde{\psi}_\xi(0, s). \tag{92}$$

In traditional calculus of variations notation, $\partial_\xi \tilde{\psi}_i(0, s)$ would be written as $\delta\psi_i(s)$ and (92) would read

$$\delta\psi_i = \left(\partial_i + \hat{\psi}_i\right)\psi_\xi. \tag{93}$$

The quantity $\tilde{\psi}_\xi$ replaces $\delta\psi$ as the fundamental field variation, and is used to construct covariant variational expressions.

1.2.3subsubsectionsection=0 . The Euler-Poincare Equations

The first variation of E in the direction of $\delta\rho = \rho\chi$, where χ is any $\mathfrak{se}(3)$-valued field on S satisfying

$$\delta\psi_k = \left(\partial_k + \hat{\psi}_k\right)\chi, \tag{94}$$

is given by

$$\begin{aligned}
\delta E_\rho(\rho\chi) &= \left\langle \frac{\delta E}{\delta\psi_k} \middle| \delta\psi_k \right\rangle_{T\mathscr{G}} && (95)\\
&= \left\langle \frac{\delta E}{\delta\psi_k} \middle| \left(\partial_k + \hat{\psi}_k\right)\chi \right\rangle_{T\mathscr{G}} && (96)\\
&= \left\langle \left(\partial_k + \hat{\psi}_k\right)^\dagger \frac{\delta E}{\delta\psi_k} \middle| \chi \right\rangle_{T\mathscr{G}} && (97)\\
&= \int_S \partial_k \left\langle \frac{\delta E}{\delta\psi_k(s)} \middle| \chi(s) \right\rangle_{\mathfrak{se}(3)} + \left\langle \left(-\partial_k + \hat{\psi}_k(s)^\dagger\right) \frac{\delta E}{\delta\psi_k(s)} \middle| \chi(s) \right\rangle_{\mathfrak{se}(3)}, && (98)
\end{aligned}$$

so, the Euler-Poincare equations are

$$\left(-\partial_k + \hat{\psi}_k(s)^\dagger\right) \frac{\delta E}{\delta\psi_k(s)} = 0 \tag{99}$$

$$\left.\frac{\delta E}{\delta\psi_k(s)}\right|_{\partial S} = 0. \tag{100}$$

We can use the structure constants (28) to express (99-100) in component form,

$$-\partial_k \frac{\partial\mathscr{L}}{\partial\psi_k^\alpha} + c_{\gamma\alpha}^\beta \psi_k^\gamma \frac{\partial\mathscr{L}}{\partial\psi_k^\beta} = 0, \text{ for } 0 \le \alpha \le 5 \tag{101}$$

$$\left.\frac{\partial \mathscr{L}}{\partial \psi_k^\alpha}\right|_{\partial S} = 0. \tag{102}$$

For the Lagrangian (76), the extremal conditions (99) and (100) read

$$\partial_j \left(B_{\alpha\beta}^{jk} \psi_k^\beta \right) + c_{\alpha\gamma}^\beta B_{\beta\mu}^{jk} \psi_j^\gamma \psi_k^\mu = 0 \tag{103}$$

Or, separating inertial and elastic stresses,

$$\partial_t \left(B_{\alpha\beta}^{tt} \psi_t^\beta \right) + c_{\alpha\gamma}^\beta B_{\beta\mu}^{tt} \psi_t^\gamma \psi_t^\mu = -\sum_{j=1}^{p} \sum_{k=1}^{p} \partial_j \left(B_{\alpha\beta}^{jk} \psi_k^\beta \right) + c_{\alpha\gamma}^\beta B_{\beta\mu}^{jk} \psi_j^\gamma \psi_k^\mu \tag{104}$$

Any nonconservative N_α or constraint C_α forces would appear on the righthand side,

$$\partial_t \left(B_{\alpha\beta}^{tt} \psi_t^\beta \right) + c_{\alpha\gamma}^\beta B_{\beta\mu}^{tt} \psi_t^\gamma \psi_t^\mu = -\sum_{j=1}^{p} \sum_{k=1}^{p} \left(\partial_j \left(B_{\alpha\beta}^{jk} \psi_k^\beta \right) + c_{\alpha\gamma}^\beta B_{\beta\mu}^{jk} \psi_j^\gamma \psi_k^\mu \right) + N_\alpha + C_\alpha \tag{105}$$

The even $\alpha = 0, 2, 4$ equations express conservation of angular momentum and the odd $\alpha = 1, 3, 5$ express the conservation of linear momentum.

1.3subsectionsection=0 . The Kirchhoff Rod ($p = 1$)

We enforce inextensibility and unshearability constraints of the Kirchhoff rod,

$$\boldsymbol{q}_0 \cdot \partial_s \boldsymbol{r}, \boldsymbol{q}_1 \cdot \partial_s \boldsymbol{r}, \boldsymbol{q}_2 \cdot \partial_s \boldsymbol{r} = \psi_s^1, \psi_s^3, \psi_s^5 = 1, 0, 0, \tag{106}$$

by adding an additional term to the Lagrangian (76),

$$\mathscr{L} = \frac{1}{2} \langle \psi | B \psi \rangle - \sum_{\mu \in \mathbb{Z}_3} \lambda_\mu \psi_s^{2\mu+1} \tag{107}$$

The constraints depend only on ψ, so, (101-102, 105) are still valid and the equations of motion are

$$\partial_t \left(B^{tt}_{\alpha\beta} \psi^\beta_t \right) + c^\beta_{\alpha\gamma} B^{tt}_{\beta\mu} \psi^\gamma_t \psi^\mu_t = -\partial_s \left(B^{ss}_{\alpha\beta} \psi^\beta_s \right) - c^\beta_{\alpha\gamma} B^{ss}_{\beta\mu} \psi^\gamma_s \psi^\mu_s + N_\alpha \tag{108}$$

$$+ \sum_{\mu \text{ odd}} \partial_s \lambda_\mu \delta^\mu_\alpha + c^\mu_{\alpha\gamma} \psi^\gamma_s \lambda_\mu \tag{109}$$

The Darboux tensor $\hat{\boldsymbol{\Omega}}$, angular velocity tensor $\hat{\boldsymbol{\omega}}$, unit tangent $\boldsymbol{e}_1$, and linear velocity vector $\boldsymbol{u}$,

$$\hat{\boldsymbol{\Omega}} = \begin{bmatrix} 0 & -\Omega_3 & \Omega_2 \\ \Omega_3 & 0 & -\Omega_1 \\ -\Omega_2 & \Omega_1 & 0 \end{bmatrix}, \quad \hat{\boldsymbol{\omega}} = \begin{bmatrix} 0 & -\omega_3 & \omega_2 \\ \omega_3 & 0 & -\omega_1 \\ -\omega_2 & \omega_1 & 0 \end{bmatrix}, \quad \boldsymbol{u} = \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}, \quad \boldsymbol{e}_1 = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} \tag{110}$$

are related to ψ by

$$\sum_{\alpha \text{ even}} \psi_1^\alpha \phi_\alpha \, \mathrm{d}s^1 = \begin{bmatrix} \hat{\boldsymbol{\Omega}} & 0 \\ 0 & 0 \end{bmatrix} \mathrm{d}s \tag{111}$$

$$\sum_{\alpha \text{ even}} \psi_0^\alpha \phi_\alpha \, \mathrm{d}s^0 = \begin{bmatrix} \hat{\boldsymbol{\omega}} & 0 \\ 0 & 0 \end{bmatrix} \mathrm{d}t \tag{112}$$

$$\sum_{\alpha \text{ odd}} \psi_1^\alpha \phi_\alpha \, \mathrm{d}s^1 = \begin{bmatrix} 0 & \boldsymbol{e}_1 \\ 0 & 0 \end{bmatrix} \mathrm{d}s \tag{113}$$

$$\sum_{\alpha \text{ odd}} \psi_0^\alpha \phi_\alpha \, \mathrm{d}s^0 = \begin{bmatrix} 0 & \boldsymbol{u} \\ 0 & 0 \end{bmatrix} \mathrm{d}t \tag{114}$$

Suppose the nonconservative force N_α is the sum of a viscous drag term which is linear in the timelike components of ψ,

$$N_\alpha = -\eta_{\alpha\beta} \psi_t^\beta, \tag{115}$$

and an active stress term A_α. Consider the limit in which inertial forces are negligible

$$B_{\alpha\beta}^{tt} \to 0. \tag{116}$$

Our equations of motion are then

$$\boxed{\begin{gathered}-\partial_s\left(B^{ss}_{\alpha\beta}\psi^\beta_s\right)-c^\beta_{\alpha\gamma}B^{ss}_{\beta\mu}\psi^\gamma_s\psi^\mu_s+\sum_{\mu\text{ odd}}\left(\partial_s\lambda_\mu\delta^\mu_\alpha-c^\mu_{\gamma\alpha}\psi^\gamma_s\lambda_\mu\right)+A_\alpha-\eta_{\alpha\beta}\psi^\beta_t=0\\ \partial_t\psi^\alpha_s-\partial_s\psi^\alpha_t=c^\alpha_{\beta\gamma}\psi^\beta_s\psi^\gamma_t\\ \psi^1_s,\psi^3_s,\psi^5_s=1,0,0\end{gathered}} \tag{117}$$

By equation (106), the centerline tangent is constant in the material frame

$$\partial_t\psi^1_1=\partial_t\psi^3_1=\partial_t\psi^5_1=0 \tag{118}$$

$$\partial_t\left(Q^T\partial_s\boldsymbol{r}\right)=0 \tag{119}$$

and the odd $\alpha=1,3,5$ components of the structure equation reduce to

$$\partial_s\psi^\alpha_t+c^\alpha_{\beta\gamma}\psi^\beta_s\psi^\gamma_t=0\text{ for }\alpha\text{ odd.} \tag{120}$$

Or in vector notation

$$\partial_s\left(Q^T\partial_t\boldsymbol{r}\right)+\boldsymbol{\Omega}\times\left(Q^T\partial_t\boldsymbol{r}\right)=0 \tag{121}$$

1.3.1subsubsectionsection=0 . Notation and some facts about Orthogonal matrices

Canonical basis for $\mathbb{R}^3$

$$\boldsymbol{e}_0,\boldsymbol{e}_1,\boldsymbol{e}_3=\begin{pmatrix}1\\0\\0\end{pmatrix},\begin{pmatrix}0\\1\\0\end{pmatrix},\begin{pmatrix}0\\0\\1\end{pmatrix} \tag{122}$$

Canonical basis for $\mathfrak{so}(3)$ (cross product matrices)

$$\hat{\mathbf{e}}_0, \hat{\mathbf{e}}_1, \hat{\mathbf{e}}_3 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ -1 & 0 & 0 \end{pmatrix}, \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{123}$$

For $\boldsymbol{u} = u_0\boldsymbol{e}_0 + u_1\boldsymbol{e}_1 + u_2\boldsymbol{e}_2$ define $\hat{\mathbf{u}} := u_0\hat{\mathbf{e}}_0 + u_1\hat{\mathbf{e}}_1 + u_2\hat{\mathbf{e}}_2$. If P is a rotation then

$$\boldsymbol{w} = P\boldsymbol{u} \Leftrightarrow \hat{\mathbf{w}} = P\hat{\mathbf{u}}P^T \tag{124}$$

Rotations about coordinate axes

$$e^{\theta\hat{\mathbf{e}}_0}, e^{\theta\hat{\mathbf{e}}_1}, e^{\theta\hat{\mathbf{e}}_2} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & \cos(\theta) & -\sin(\theta) \\ 0 & \sin(\theta) & \cos(\theta) \end{pmatrix}, \begin{pmatrix} \cos(\theta) & 0 & \sin(\theta) \\ 0 & 1 & 0 \\ -\sin(\theta) & 0 & \cos(\theta) \end{pmatrix}, \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{125}$$

If $Q \in SO(3)$ transforms $\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2$ to $\boldsymbol{q}_0 = Q\boldsymbol{e}_0, \boldsymbol{q}_1 = Q\boldsymbol{e}_1, \boldsymbol{q}_2 = Q\boldsymbol{e}_2$ then rotations about $\boldsymbol{q}_i$'s are given by

$$e^{\theta\hat{\mathbf{q}}_i} = Qe^{\theta\hat{\mathbf{e}}_i}Q^{-1} \tag{126}$$

1.4subsectionsection=0 . Framed curves

A curve $\boldsymbol{r}$ with adapted frame $\boldsymbol{q}_0 = \boldsymbol{r}', \boldsymbol{q}_1, \boldsymbol{q}_2$ can be used to define an $SE(3)$-valued matrix

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{q}_0 \boldsymbol{e}_0^T + \boldsymbol{q}_1 \boldsymbol{e}_1^T + \boldsymbol{q}_2 \boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{127}$$

The *shape* of the framed curve, or characteristics invariant under rigid transformation, can be described by a curve through the special Euclidean algebra $\mathfrak{se}(3)$

$$\psi = \rho^{-1}\rho' = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \tag{128}$$

Requiring $\boldsymbol{q}_0 = \boldsymbol{r}'$ is equivalent to $\boldsymbol{U} = \boldsymbol{e}_0$. The framed curve can be recovered from ψ by solving the initial value problem

$$\rho' = \rho\psi, \ \rho(0) = \rho_0 \tag{129}$$

The solution to (129)

$$\rho(s) = \rho_0 Oexp(\psi, s) \tag{130}$$

is given in terms of the ordered exponential

$$Oexp(\psi, s) = \prod_{\xi=0}^{\xi=s} e^{\psi(\xi)\mathrm{d}\xi} = \lim_{n\to\infty} e^{\psi(\xi_0)\mathrm{d}\xi} e^{\psi(\xi_1)\mathrm{d}\xi} \dots e^{\psi(\xi_{n-1})\mathrm{d}\xi} \tag{131}$$

where $0 = \xi_0 < \xi_1 < \ldots < \xi_{n-1} = s$ is a partition of the interval $[0, s]$. The related problem

$$\rho' = \underline{\psi}\rho \tag{132}$$

$$\rho(0) = \rho_0, \tag{133}$$

where $\underline{\psi} = \rho\psi\rho^{-1}$, can be solved by reversing the above path ordering

$$\rho(s) = \lim_{n\to\infty} e^{\underline{\psi}(\xi_{n-1})\mathrm{d}\xi} e^{\underline{\psi}(\xi_{n-2})\mathrm{d}\xi} \ldots e^{\underline{\psi}(\xi_0)\mathrm{d}\xi}\rho_0 \tag{134}$$

Suppose $\boldsymbol{r}$ has Frenet-Serret frame $\boldsymbol{T}, \boldsymbol{N}, \boldsymbol{B}$ curvature κ and torsion τ.

$$\mu = \begin{pmatrix} F & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{T}\boldsymbol{e}_0^T + \boldsymbol{N}\boldsymbol{e}_1^T + \boldsymbol{B}\boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{135}$$

$$\mu^{-1}\mu' = \begin{pmatrix} \tau\hat{\mathbf{e}}_0 + \kappa\hat{\mathbf{e}}_2 & \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \tag{136}$$

Any two adapted frames are related by rotation about curve tangent. As a consequence, we find the following

$$Q = e^{-\phi\hat{\mathbf{T}}}F = Fe^{-\phi\hat{\mathbf{e}}_0} \tag{137}$$

$$Q\boldsymbol{U} = F\boldsymbol{U} = \boldsymbol{r}' \tag{138}$$

$$\phi(s) = \phi_0 + \int_0^s \tau - \Omega_0 \,\mathrm{d}s \tag{139}$$

$$\boldsymbol{\Omega} = \Omega_0 \boldsymbol{e}_0 + \boldsymbol{\Omega}_\perp = \Omega_0 \boldsymbol{e}_0 + \Omega_1 \boldsymbol{e}_1 + \Omega_2 \boldsymbol{e}_2 \tag{140}$$

$$\boldsymbol{\Omega}_\perp = \kappa e^{\phi \hat{\mathbf{e}}_0} \boldsymbol{e}_2 \tag{141}$$

$$Q\boldsymbol{\Omega}_\perp = \boldsymbol{r}' \times \boldsymbol{r}'' = \kappa \boldsymbol{B} \tag{142}$$

$$\kappa = |\boldsymbol{\Omega}_\perp| \tag{143}$$

$$\tau = \Omega_0 + \frac{\boldsymbol{e}_0 \cdot (\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}'_\perp)}{|\boldsymbol{\Omega}_\perp|^2} = \Omega_0 + \phi' \tag{144}$$

1.5subsectionsection=0 . Time dependent framed curves

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix}, \ \psi = \rho^{-1}\,\mathrm{d}\rho \tag{145}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \mathrm{d}s + \begin{pmatrix} \hat{\boldsymbol{\omega}} & \boldsymbol{u} \\ 0 & 0 \end{pmatrix} \mathrm{d}t \tag{146}$$

$$Q\boldsymbol{U} = \boldsymbol{r}' \tag{147}$$

Angular $\boldsymbol{\omega}$ and linear $\boldsymbol{u}$ velocities appear naturally as timelike analogues of the twist/curvature $\boldsymbol{\Omega}$ and curve tangent $\boldsymbol{U}$. Commutativity of partial derivatives requires ψ satisfy an additional *structure equation*

$$\rho^{-1}\,\mathrm{d}^2\rho = \mathrm{d}\psi + \psi \wedge \psi = 0. \tag{148}$$

The structure equation (148) together with the constraint (147) imply

$$\dot{\Omega} = \boldsymbol{\omega}' + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{149}$$

$$\boldsymbol{\omega} \times \boldsymbol{e}_0 = \boldsymbol{u}' + \boldsymbol{\Omega} \times \boldsymbol{u} \tag{150}$$

Note that imply the familiar consitency conditions found in [**?**]

$$\dot{\Omega}_0 = \omega_0' + \boldsymbol{r}' \cdot (\boldsymbol{r}'' \times \dot{\boldsymbol{r}}') \tag{151}$$

$$\boldsymbol{r}' \cdot \dot{\boldsymbol{r}}' = 0 \tag{152}$$

Components of ψ are related to inertial frame derivatives of $\boldsymbol{r}$ by

$$\boldsymbol{\Omega} = \Omega_0 \boldsymbol{e}_0 + \Omega_1 \boldsymbol{e}_1 + \Omega_2 \boldsymbol{e}_2 \tag{153}$$

$$Q\boldsymbol{\Omega}_\perp = \boldsymbol{r}' \times \boldsymbol{r}'' = \kappa \boldsymbol{B} \tag{154}$$

$$Q\boldsymbol{U} = \boldsymbol{r}' \tag{155}$$

$$\boldsymbol{\omega} = \omega_0 \boldsymbol{e}_0 + \omega_1 \boldsymbol{e}_1 + \omega_2 \boldsymbol{e}_2 \tag{156}$$

$$Q\boldsymbol{\omega}_\perp = \boldsymbol{r}' \times \dot{\boldsymbol{r}}' \tag{157}$$

$$\boldsymbol{u} = u_0 \boldsymbol{e}_0 + u_1 \boldsymbol{e}_1 + u_2 \boldsymbol{e}_2 \tag{158}$$

$$Q\boldsymbol{u} = \dot{\boldsymbol{r}} \tag{159}$$

1.5.1subsubsectionsection=0 . $U(1)$ gauge transformations

When representing a filament with circular cross-section by a framed curve, the initial direction chosen for the transverse frame vectors $\boldsymbol{q}_1$, $\boldsymbol{q}_2$ is equivalent to fixing a gauge. The gauge group $U(1)$ acts by rotating the frame at each point along the curve by a constant angle about the curve tangent $\boldsymbol{q}_0 = \boldsymbol{r}'$. Gauge invariants include the centerline $\boldsymbol{r}$ and all associated geometric quatities (e.g. curvature, tor-

sion, and Frenet-Serret frame), as well as the twist Ω_0.

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix}, \ = \rho^{-1}\,\mathrm{d}\rho \tag{160}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{U} \\ 0 & 0 \end{pmatrix} \mathrm{d}s + \begin{pmatrix} \hat{\boldsymbol{\omega}} & \boldsymbol{u} \\ 0 & 0 \end{pmatrix} \mathrm{d}t \tag{161}$$

Gauge transformations can be represented by $SE(3)$-valued matrices

$$\gamma = \begin{pmatrix} e^{\theta\hat{\mathbf{e}}_0} & 0 \\ 0 & 1 \end{pmatrix} \tag{162}$$

whose action on ρ is defined by

$$\begin{aligned} \rho \mapsto \rho\gamma^{-1} &= \begin{pmatrix} Qe^{-\theta\hat{\mathbf{e}}_0} & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} e^{-\theta\hat{\mathbf{q}}_0}Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \end{aligned} \tag{163}$$

Under a change of guage, ψ transforms as

$$\psi \mapsto \gamma\psi\gamma^{-1}. \tag{164}$$

Or, in terms of its' components

$$\boldsymbol{\Omega} \mapsto \Omega_0 \boldsymbol{e}_0 + e^{\theta \hat{\mathbf{e}}_0} \boldsymbol{\Omega}_\perp \tag{165}$$

$$\boldsymbol{U} \mapsto \boldsymbol{U} \tag{166}$$

$$\boldsymbol{\omega} \mapsto \omega_0 \boldsymbol{e}_0 + e^{\theta \hat{\mathbf{e}}_0} \boldsymbol{\omega}_\perp \tag{167}$$

$$\boldsymbol{u} \mapsto u_0 \boldsymbol{e}_0 + e^{\theta \hat{\mathbf{e}}_0} \boldsymbol{u}_\perp \tag{168}$$

1.5.2subsubsectionsection=0 . Symmetries

Let ρ describe an arclength parameterized $\boldsymbol{r}(s)$ and its' Frenet-Serret frame $\boldsymbol{T}, \boldsymbol{N}, \boldsymbol{B}$ with curvature κ and torsion τ.

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{T}\boldsymbol{e}_0^T + \boldsymbol{N}\boldsymbol{e}_1^T + \boldsymbol{B}\boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{169}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau \hat{\mathbf{e}}_0 + \kappa \hat{\mathbf{e}}_2 & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} \tag{170}$$

The curvature κ and torsion τ are even functions of s iff the curve is invariant under rotation by π about an axis which intersects the curve at $\boldsymbol{r}(0)$ and which is perpendicular to $\boldsymbol{T}(0)$. If the curvature is nonvanishing at $s = 0$, then the axis of rotation is

parallel to $\boldsymbol{n} = \boldsymbol{N}(0)$.

$$\psi(-s) = \psi(s) \tag{171}$$

$$\Leftrightarrow \tag{172}$$

$$\rho(-s) = \rho_0 \gamma \rho_0^{-1} \rho(s) \gamma^{-1} \tag{173}$$

$$= \begin{pmatrix} e^{\pi \hat{\mathbf{N}}(0)} Q(s) e^{\pi \hat{\mathbf{e}}_1} & \boldsymbol{r}(0) + e^{\pi \hat{\mathbf{N}}(0)} (\boldsymbol{r}(s) - \boldsymbol{r}(0)) \\ 0 & 1 \end{pmatrix} \tag{174}$$

where γ is given by

$$\gamma = \begin{pmatrix} e^{\pi \hat{\mathbf{e}}_1} & 0 \\ 0 & 1 \end{pmatrix} \tag{175}$$

Proof:

$$\rho = \begin{pmatrix} Q & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} \boldsymbol{T}\boldsymbol{e}_0^T + \boldsymbol{N}\boldsymbol{e}_1^T + \boldsymbol{B}\boldsymbol{e}_2^T & \boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{176}$$

$$\psi = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} \tau \hat{\mathbf{e}}_0 + \kappa \hat{\mathbf{e}}_2 & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} \tag{177}$$

To see this, first, assume $\rho_0 = 1$, so that $\boldsymbol{T}(0) = \boldsymbol{e}_0$, $\boldsymbol{N}(0) = \boldsymbol{e}_1$, $\boldsymbol{B}(0) = \boldsymbol{e}_2$ and $\boldsymbol{r}(0) = 0$.

$$\rho(s) = \prod_{\xi=0}^{\xi=s} e^{\psi(\xi) \mathrm{d}\xi} \tag{178}$$

$$\rho(-s) = \prod_{\xi=0}^{\xi=-s} e^{\psi(\xi)\mathrm{d}\xi} = \prod_{\xi=0}^{\xi=s} e^{-\psi(-\xi)\mathrm{d}\xi} = \prod_{\xi=0}^{\xi=s} e^{-\psi(\xi)\mathrm{d}\xi} \tag{179}$$

Where the last equality uses $\psi(-s) = \psi(s)$. If we define

$$\gamma = \begin{pmatrix} e^{\pi\hat{\mathbf{e}}_1} & 0 \\ 0 & 1 \end{pmatrix} = \begin{pmatrix} -\boldsymbol{e}_0\boldsymbol{e}_0^T + \boldsymbol{e}_1\boldsymbol{e}_1^T - \boldsymbol{e}_2\boldsymbol{e}_2^T & 0 \\ 0 & 1 \end{pmatrix} \tag{180}$$

then we have

$$\gamma\psi\gamma^{-1} = \begin{pmatrix} e^{\pi\hat{\mathbf{e}}_1}\left(\tau\hat{\mathbf{e}}_0 + \kappa\hat{\mathbf{e}}_2\right)e^{-\pi\hat{\mathbf{e}}_1} & e^{\pi\hat{\mathbf{e}}_1}\boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = \begin{pmatrix} -\tau\hat{\mathbf{e}}_0 - \kappa\hat{\mathbf{e}}_2 & -\boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix} = -\psi \tag{181}$$

$$\rho(-s) = \prod_{\xi=0}^{\xi=s} e^{\gamma\psi(\xi)\gamma^{-1}\mathrm{d}\xi} = \gamma\left(\prod_{\xi=0}^{\xi=s} e^{\psi(\xi)\mathrm{d}\xi}\right)\gamma^{-1} = \gamma\rho(s)\gamma^{-1} \tag{182}$$

$$\gamma\rho\gamma^{-1} = \begin{pmatrix} e^{\pi\hat{\mathbf{e}}_1}Qe^{-\pi\hat{\mathbf{e}}_1} & e^{\pi\hat{\mathbf{e}}_1}\boldsymbol{r} \\ 0 & 1 \end{pmatrix} \tag{183}$$

More generally, if $\rho(s) = \rho_0 Oexp(\psi, s)$, then $\rho_0^{-1}\rho$ satisfies $\rho_0^{-1}\rho(-s) = \gamma\rho_0^{-1}\rho(s)\gamma^{-1}$

$$\rho(-s)\rho_0\gamma\rho_0^{-1}\rho\gamma^{-1} = \rho(0)\begin{pmatrix} e^{\pi\hat{\mathbf{e}}_1} & 0 \\ 0 & 1 \end{pmatrix}\rho(0)^{-1}\rho(s)\begin{pmatrix} e^{-\pi\hat{\mathbf{e}}_1} & 0 \\ 0 & 1 \end{pmatrix} \tag{184}$$

□

1.5.3subsubsectionsection=0 . Evolution equations – inertial frame

If Ω_0, κ, τ are even, then there exists a gauge in which Ω_1 is odd and Ω_2 is even. If m and f are also odd at $t = 0$, then this symmetry is conserved. Boundary conditions at $s = 0$ are given by

$$\Omega_0' = 0 \tag{185}$$

$$\Omega_1 = 0, \Omega_1'' = 0 \tag{186}$$

$$\Omega_2' = 0, \Omega_2''' = 0 \tag{187}$$

$$\lambda' = 0 \tag{188}$$

$$Q\boldsymbol{\Omega} = \Omega_0 \boldsymbol{r}' + \boldsymbol{r}' \times \boldsymbol{r}'' \tag{189}$$

$$Q\boldsymbol{\omega} = \omega_0 \boldsymbol{r}' + \boldsymbol{r}' \times \dot{\boldsymbol{r}}' \tag{190}$$

$$Q\boldsymbol{U} = \boldsymbol{r}' \tag{191}$$

$$Q\boldsymbol{u} = \dot{\boldsymbol{r}} \tag{192}$$

Structure equation

$$\dot{\Omega}_0 = \omega_0' + \boldsymbol{r}' \cdot (\boldsymbol{r}'' \times \dot{\boldsymbol{r}}') \tag{193}$$

$$\boldsymbol{r}' \cdot \dot{\boldsymbol{r}}' = 0 \tag{194}$$

Force and moment balance

$$\left(\beta_\parallel \Omega_0 \boldsymbol{r}' + \beta_\perp \boldsymbol{r}' \times \boldsymbol{r}''\right)' + \boldsymbol{r}' \times \boldsymbol{\Lambda} + m\boldsymbol{r}' - \zeta_r \omega_0 \boldsymbol{r}' = 0 \tag{195}$$

$$\boldsymbol{\Lambda}' + f\boldsymbol{r}' - \zeta \dot{\boldsymbol{r}} = 0 \tag{196}$$

where $\zeta = \zeta_\parallel \boldsymbol{r}' \boldsymbol{r}'^T + \zeta_\perp (1 - \boldsymbol{r}' \boldsymbol{r}'^T)$. ... ...

Moment balance gives

$$\omega_0 = \frac{\beta_\parallel}{\zeta_r}\Omega_0' + \frac{1}{\zeta_r}m \tag{197}$$

$$\boldsymbol{\Lambda} = \lambda \boldsymbol{r}' + \boldsymbol{r}' \times \left(\beta_\parallel \Omega_0 \boldsymbol{r}'' + \beta_\perp \boldsymbol{r}' \times \boldsymbol{r}'''\right) \tag{198}$$

Dotting force balance with $\boldsymbol{r}'$ and taking a derivative wrt s and using $\boldsymbol{r}'\dot{\boldsymbol{r}}' = 0$ gives

$$\boldsymbol{r}' \cdot \boldsymbol{\Lambda}' + f - \zeta_\parallel \boldsymbol{r}' \cdot \dot{\boldsymbol{r}} = 0 \tag{199}$$

$$(\boldsymbol{r}' \cdot \boldsymbol{\Lambda}')' + f' - \zeta_\parallel \boldsymbol{r}'' \cdot \dot{\boldsymbol{r}} = 0 \tag{200}$$

Dotting force balance with $\boldsymbol{r}''$, solving for $\boldsymbol{r}'' \cdot \dot{\boldsymbol{r}}$, and using the above gives the tension equation

$$\boldsymbol{r}'' \cdot \dot{\boldsymbol{r}} = \frac{1}{\zeta_\perp}\boldsymbol{r}'' \cdot \boldsymbol{\Lambda}' \tag{201}$$

$$(\boldsymbol{r}' \cdot \boldsymbol{\Lambda}')' + f' - \frac{\zeta_\parallel}{\zeta_\perp}\boldsymbol{r}'' \cdot \boldsymbol{\Lambda}' = 0 \tag{202}$$

$$\lambda'' - \frac{\zeta_\parallel}{\zeta_\perp}\kappa^2\lambda + f' + \frac{1}{2}\beta_\perp(\kappa^2)'' + \frac{\zeta_\parallel}{\zeta_\perp}\kappa^2\tau\left(\beta_\parallel\Omega_0 - \beta_\perp\tau\right) + \frac{\zeta_\parallel\beta_\perp}{\zeta_\perp}\kappa\kappa'' = 0 \tag{203}$$

Torsion can be eliminated using

$$\Omega_0\kappa^2\tau = \kappa^2\Omega_0^2 + \Omega_0\boldsymbol{e}_0 \cdot \left(\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}_\perp'\right) \tag{204}$$

$$\kappa\kappa'' - \kappa^2\tau^2 = \boldsymbol{\Omega}_\perp \cdot \boldsymbol{\Omega}_\perp'' - \kappa^2\Omega_0^2 - 2\Omega_0\boldsymbol{e}_0 \cdot \left(\boldsymbol{\Omega}_\perp \times \boldsymbol{\Omega}_\perp'\right) \tag{205}$$

$$\lambda'' - \frac{\zeta_\parallel}{\zeta_\perp}\kappa^2\lambda + f' + \frac{1}{2}\beta_\perp(\kappa^2)'' \tag{206}$$

$$+\frac{\zeta_\parallel\beta_\parallel}{\zeta_\perp}\left(\kappa^2\Omega_0^2 + \Omega_0\boldsymbol{e}_0\cdot\left(\boldsymbol{\Omega}_\perp\times\boldsymbol{\Omega}_\perp'\right)\right) \tag{207}$$

$$+\frac{\zeta_\parallel\beta_\perp}{\zeta_\perp}\left(\boldsymbol{\Omega}_\perp\cdot\boldsymbol{\Omega}_\perp'' - \kappa^2\Omega_0^2 - 2\Omega_0\boldsymbol{e}_0\cdot\left(\boldsymbol{\Omega}_\perp\times\boldsymbol{\Omega}_\perp'\right)\right) \tag{208}$$

$$= 0 \tag{209}$$

2sectionsection=0 . Swimming P. Mirabilis

This section models swimming P. Mirabilis using (117). Notation and a few important equations are in (1.2). Dimensional constants and material properties of P. Mirabilis are discussed in (1.3). Evolution equations for components of the Darboux vector are described in (1.4). Time evolution of the active force and moment densities is discussed in (1.5). Section (1.6) covers time evolution and integration of centerline and material frame. Boundary conditions can be found in (1.7). The full system of nondimensionalized equations of motion for the swimming P. Mirabilis are in (1.8). Solutions to the equations in (1.8) and swimming trajectories are analyzed in (4). Section (**??**) has some explicit expressions for some quantities of interest produced with Sympy. Details about the numerical methods are in (5).

2.1subsectionsection=0 . mathematical model

... The cell is modeled as a semiflexible filament with active force and moment distributions. The standard

The cell is modeled as a semi-flexible filament (Kirchhoff rod) [**?**] with active force $\underline{f}_a = f_a(s,t)\underline{e}_0$ and moment $\underline{m}_a = m_a(s,t)\underline{e}_0$ distributions, where $\underline{e}_0$ is the local tangent vector, s is the arc-length, and t is time. Viscous stresses are treated using resistive-force theory $f_v = -c_\parallel u_\parallel - c_\perp u_\perp$ with an additional viscous moment $m_v = -c_r\omega_\parallel$ proportional to the tangential angular velocity $\omega_\parallel$.

$$\delta\int \frac{1}{2}B_\parallel|\Omega_\parallel|^2 + \frac{1}{2}B_\perp|\Omega_\perp|^2 + \lambda\cdot(Q^{-1}\partial_s r - e_0)\,\mathrm{d}s = \delta W_a + \delta W_v.$$

Force and moment balance together with the commutation relation for $SE(3)$ result in a system of nonlinear PDEs which describe the evolution of the cell:

$$\partial_t\Omega_\parallel - \left[\frac{\beta_\parallel}{\eta_r} + \frac{\beta_\parallel - \beta_\perp}{\eta_\perp}|\Omega_\perp|^2\right]\partial_s^2\Omega_\parallel = G_\parallel(\Omega_\parallel, \partial_s\Omega_\parallel, \Omega_\perp\ldots, \partial_s^3\Omega_\perp, \lambda, \partial_s m_a),$$

$$\partial_t \Omega_\perp + \frac{\beta_\perp}{\eta_\perp}\partial_s^4 \Omega_\perp = G_\perp(\Omega_\parallel, \ldots, \partial_s^2 \Omega_\parallel, \Omega_\perp \ldots, \partial_s^3 \Omega_\perp, \lambda, \ldots, \partial_s^2 \lambda, f_a, \partial_s f_a, m_a),$$

$$\partial_s^2 \lambda - \frac{1}{\eta_\perp}|\Omega_\perp|^2 \lambda = G_\lambda(\Omega_\parallel, \Omega_\perp \ldots, \partial_s^2 \Omega_\perp, \partial_s f_a).$$

The system is governed by four dimensionless parameters:

$$\beta_\parallel = B_\parallel/(f_a^* L^3), \beta_\perp = B_\perp/(f_a^* L^3), \eta_\parallel = c_\perp/c_\parallel, \eta_r = c_r/(c_\parallel L^2)$$

2.2subsectionsection=0 . Notation, force/moment balance, and constraints

Unless otherwise stated, all vector/tensors are represented by component matrices in the material frame.

- Canonical basis for $\mathbb{R}^3$

$$\boldsymbol{e}_0, \boldsymbol{e}_1, \boldsymbol{e}_2 = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{210}$$

- Material frame — $Q \in SO(3)$

$$Q = Q_{ij}\boldsymbol{e}_i \boldsymbol{e}_j^T = \boldsymbol{q}_j \boldsymbol{e}_j^T \tag{211}$$

- Centerline position vector — $\boldsymbol{x} \in \mathbb{R}^3$

$$\boldsymbol{x} = x_i \boldsymbol{e}_i \tag{212}$$

- Inextensibility/unshearability constraint

$$\partial_s (Q\boldsymbol{x}) = Q\boldsymbol{e}_0 \tag{213}$$

- Cross-section pose $\rho \in SE(3)$

$$\rho = \begin{pmatrix} Q & Q\boldsymbol{x} \\ 0 & 1 \end{pmatrix} \tag{214}$$

- Darboux vector — $\boldsymbol{\Omega} = \Omega_i \boldsymbol{e}_i$

$$\partial_s Q\boldsymbol{e}_k = Q(\boldsymbol{\Omega} \times \boldsymbol{e}_k) \tag{215}$$

  - Twist — Ω_0
  - Curvature — Ω_1, Ω_2

- Translational velocity — $\boldsymbol{u} = u_i \boldsymbol{e}_i = Q^T \partial_t (Q\boldsymbol{x})$
- Angular velocity — $\boldsymbol{\omega} = \omega_i \boldsymbol{e}_i$

$$\partial_t Q\boldsymbol{e}_k = Q(\boldsymbol{\omega} \times \boldsymbol{e}_k) \tag{216}$$

- s,t-generators, Pfaffian constraints

$$\psi_s = \begin{pmatrix} \hat{\boldsymbol{\Omega}} & \boldsymbol{e}_0 \\ 0 & 0 \end{pmatrix}, \qquad \psi_t = \begin{pmatrix} \hat{\boldsymbol{\omega}} & \boldsymbol{u} \\ 0 & 0 \end{pmatrix} \tag{217}$$

$$\partial_s \rho = \rho \psi_s, \qquad \partial_t \rho = \rho \psi_t \tag{218}$$

Note that these constraints are equivalent to

$$\partial_s Q = Q\hat{\boldsymbol{\Omega}}, \qquad \partial_t Q = Q\hat{\boldsymbol{\omega}} \tag{219}$$

$$\partial_s (Q\boldsymbol{x}) = Q\boldsymbol{e}_1, \qquad \partial_t (Q\boldsymbol{x}) = Q\boldsymbol{u} \tag{220}$$

- Compatibility conditions

$$\partial_t \boldsymbol{\Omega} = \partial_s \boldsymbol{\omega} + \boldsymbol{\Omega} \times \boldsymbol{\omega} \tag{221}$$

$$\partial_s \boldsymbol{u} + \boldsymbol{\Omega} \times \boldsymbol{u} = \boldsymbol{\omega} \times \boldsymbol{e}_0 \tag{222}$$

- Internal force on s from $(s, L]$ section — $\boldsymbol{f} = f_i \boldsymbol{e}_i = \lambda \boldsymbol{e}_0 + f_1 \boldsymbol{e}_1 + f_2 \boldsymbol{e}_2$
- Stiffness tensor and internal moment on s from $(s, L]$ section — $B = B_\parallel \boldsymbol{e}_1 \boldsymbol{e}_1^T + B_\perp \left(\boldsymbol{e}_2 \boldsymbol{e}_2^T + \boldsymbol{e}_3 \boldsymbol{e}_3^T\right)$

$$\boldsymbol{m} = B\boldsymbol{\Omega} \tag{223}$$

- Translational drag tensor and force per length — $c = c_\parallel \boldsymbol{e}_1 \boldsymbol{e}_1^T + c_\perp \left(\boldsymbol{e}_2 \boldsymbol{e}_2^T + \boldsymbol{e}_3 \boldsymbol{e}_3^T\right)$

$$\boldsymbol{f}_v = -c\boldsymbol{u} \tag{224}$$

- Axial drag and moment per length — $c_r \in \mathbb{R}$

$$\boldsymbol{m}_v = -c_r \omega_1 \boldsymbol{e}_1 \tag{225}$$

- Active force and moment per length — f_a, m_a

$$\boldsymbol{f}_a = f_a \boldsymbol{e}_1, \qquad \boldsymbol{m}_a = m_a \boldsymbol{e}_1 = \chi f_a \boldsymbol{e}_1 \tag{226}$$

- Trajectories of the body are governed by the field equations (117) which ex-

press moment balance ($\alpha = 0,2,4$), force balance ($\alpha = 1,3,5$), compatibility conditions, and inextensibility/unshearability. They are equivalent to the vector equations

$$
\begin{aligned}
B\partial_s \boldsymbol{\Omega} + \boldsymbol{\Omega} \times B\boldsymbol{\Omega} + \boldsymbol{e}_1 \times \boldsymbol{f} - c_r \omega_1 \boldsymbol{e}_1 + m_a \boldsymbol{e}_1 &= 0 \\
\partial_s \boldsymbol{f} + \boldsymbol{\Omega} \times \boldsymbol{f} - c_\parallel \boldsymbol{u}_\parallel - c_\perp \boldsymbol{u}_\perp + f_a \boldsymbol{e}_1 &= 0 \\
\partial_s \partial_t Q &= \partial_t \partial_s Q \\
\partial_s \partial_t (Q\boldsymbol{x}) &= \partial_t \partial_s (Q\boldsymbol{x}) \\
\partial_s (Q\boldsymbol{x}) &= Q\boldsymbol{e}_1
\end{aligned}
\tag{227}
$$

If we neglect the active force and moment, then (227) are the standard equations used to describe motion of a Kirchhoff rod in a viscous fluid [1], [2].

2.3subsectionsection=0 . Units and nondimensional parameters

Let the body length L, characteristic swimming speed U, and parallel drag coefficient $c_\parallel$ define our fundamental system of units.

- Body dimensions – $[L] = [\ell] = \text{ m}$
  - $L \sim 10^{-5}$ m
  - Body diameter – $\sim 10^{-6}$ m
- Characteristic swimming speed – $[U] = \frac{\text{m}}{\text{s}}$
  - $U \approx 1.5 \times 10^{-5} \frac{\text{m}}{\text{s}}$
- Elastic stresses
  - Twisting/bending stiffness – $[B_\parallel] = [B_\perp] = \text{ N} \cdot \text{m}^2$
    - $B_\parallel \approx$???
    - $B_\perp \approx 5 \times 10^{-22} - 1.4 \times 10^{-20}$ N $\cdot$ m^2

- Elastic moment – $[\boldsymbol{m}] = \mathrm{N}\cdot\mathrm{m}$
  - $\boldsymbol{m} = B_\parallel \boldsymbol{\Omega}_\parallel + B_\perp \boldsymbol{\Omega}_\perp$
- Elastic force – $[\boldsymbol{f}_\perp] = \mathrm{N}$
  - $\boldsymbol{f}_\perp = B_\perp \boldsymbol{e}_1 \times \boldsymbol{\Omega}_\parallel + (B_\parallel - B_\perp)\Omega_1 \boldsymbol{\Omega}_\perp$

- Viscous Stresses
  - Translational drag coefficients – $[c_\parallel] = [c_\perp] = \frac{\mathrm{N\cdot s}}{\mathrm{m}^2}$
    - $c_\parallel \approx 4.95 \times 10^{-2} \frac{\mathrm{N\cdot s}}{\mathrm{m}^2}$
    - $c_\perp \approx 7.83 \times 10^{-2} \frac{\mathrm{N\cdot s}}{\mathrm{m}^2}$
  - Rotational drag coefficient – $[c_r] = \mathrm{N}\cdot\mathrm{s}$
    - $c_r \approx 1.25 \times 10^{-14}\ \mathrm{N}\cdot\mathrm{s}$

- Active stresses
  - Characteristic force density – $[f_a^*] = \frac{\mathrm{N}}{\mathrm{m}}$
    - $f_a^* \approx c_\parallel U = 7.43 \times 10^{-7}$
  - Characteristic moment density – $[m_a^*] = \mathrm{N}$
    - $m_a^* \approx 0$
  - Diffusion coefficient – $[\sigma] = \frac{\mathrm{m}^2}{s}$
    - $\sigma \approx UL \times 10^{-2} \approx 1.5^{-12} \frac{\mathrm{m}^2}{s}$

Nondimensional stuff:

- Drag ratios

$$\eta_\perp = \frac{c_\perp}{c_\parallel} \approx 1.585 \tag{228}$$

$$\eta_r = \frac{c_r}{c_\parallel L^2} \approx 2.53 \times 10^{-3} \tag{229}$$

- Elastic-active ratios

$$\beta_{\parallel} = \frac{B_{\parallel}}{f_a^* L^3} \approx ??? \tag{230}$$

$$\beta_{\perp} = \frac{B_{\perp}}{f_a^* L^3} \approx .673 - 18.8 \tag{231}$$

2.4subsectionsection=0 . Twist/curvature evolution and tension

The field equations (117) can be reduced to a system of four coupled equations in the twist Ω_1, curvatures Ω_2, Ω_3, and tension λ

$$\partial_t \Omega_1 + \left[\frac{B_{\perp} - B_{\parallel}}{c_{\perp}} |\boldsymbol{\Omega}_{\perp}|^2 - \frac{B_{\parallel}}{c_r} \right] \partial_s^2 \Omega_1 = G_1 \left(\Omega_1, \partial_s \Omega_1, \boldsymbol{\Omega}_{\perp}, \ldots, \partial_s^3 \boldsymbol{\Omega}_{\perp}, \lambda, \partial_s m_a \right) \tag{232}$$

$$\partial_t \boldsymbol{\Omega}_{\perp} + \frac{B_{\perp}}{c_{\perp}} \partial_s^4 \boldsymbol{\Omega}_{\perp} + \frac{B_{\perp} - B_{\parallel}}{c_{\perp}} \boldsymbol{e}_1 \times \boldsymbol{\Omega}_{\perp} \partial_s^3 \Omega_1 = G_{\perp} \left(\Omega_1, \ldots, \partial_s^2 \Omega_1, \boldsymbol{\Omega}_{\perp}, \ldots, \partial_s^3 \boldsymbol{\Omega}_{\perp}, \lambda, \ldots, \partial_s^2 \lambda, f_a, \partial_s f_a, m_a \right) \tag{233}$$

$$\partial_s^2 \lambda - \frac{c_{\parallel}}{c_{\perp}} |\boldsymbol{\Omega}_{\perp}|^2 \lambda = G_{\lambda} \left(\Omega_1, \boldsymbol{\Omega}_{\perp}, \ldots, \partial_s^2 \boldsymbol{\Omega}_{\perp}, \partial_s f_a \right) \tag{234}$$

To show this, we need to express $f_2, f_3, \omega_1, \omega_2, \omega_3, u_1, u_2, u_3$ in terms of $\Omega_1, \Omega_2, \Omega_3, \lambda$. From force/moment balance (227) and the translational part of the structure equation (222), we find

$$\boldsymbol{f} = \lambda \boldsymbol{e}_1 + \boldsymbol{e}_1 \times (B \partial_s \boldsymbol{\Omega} + \boldsymbol{\Omega} \times B \boldsymbol{\Omega}) \tag{235}$$

$$\boldsymbol{u} = c^{-1} \left(\partial_s \boldsymbol{f} + \boldsymbol{\Omega} \times \boldsymbol{f} + f_a \boldsymbol{e}_1 \right) \tag{236}$$

$$\boldsymbol{\omega} = \frac{1}{c_r} \left[\boldsymbol{e}_1 \cdot (B \partial_s \boldsymbol{\Omega} + \boldsymbol{\Omega} \times B \boldsymbol{\Omega}) + m_a \right] \boldsymbol{e}_1 + \boldsymbol{e}_1 \times (\partial_s \boldsymbol{u} + \boldsymbol{\Omega} \times \boldsymbol{u}) . \tag{237}$$

Substituting (235) into (236) and then the result into (237) eliminates the desired variables. Finally, the tension ODE (234) comes from $\boldsymbol{e}_1$ component of (222) and

the curvature PDE's (232-233) come from (221).

2.5subsectionsection=0 . Active force/moment evolution

$$\partial_t f_a = \frac{f_a^*}{L}\left[1-\left(\frac{f_a}{f_a^*}\right)^2\right]u_1 + \sigma\partial_s^2 f_a \tag{238}$$

$$\partial_s f_a\big|_{s=0,L} = 0 \tag{239}$$

$$m_a = \frac{m_a^*}{f_a^*} f_a \tag{240}$$

The active power per unit length is

$$\partial_s P_{active} = f_a u_1 + m_a \omega_1 \tag{241}$$

$$= \frac{B_\perp}{c_\parallel}\boldsymbol{\Omega}_\perp \cdot \partial_s \boldsymbol{\Omega}_\perp f_a + \frac{1}{c_\parallel}\partial_s \lambda f_a + \frac{1}{c_\parallel} f_a^2 + \frac{B_\parallel}{c_r}\partial_s \Omega_1 m_a + \frac{1}{c_r} m_a^2 \tag{242}$$

$$= \partial_s\left(\frac{B_\perp}{2\mu_\parallel}|\boldsymbol{\Omega}_\perp|^2 + \frac{1}{c_\parallel}\lambda + \frac{m_a^* B_\parallel}{f_a^* c_r}\Omega_1\right) f_a + \left(\frac{1}{c_\parallel} + \frac{m_a^{*2}}{f_a^{*2} c_r}\right) f_a^2 \tag{243}$$

2.6subsectionsection=0 . Pose integration

To integrate the frame and centerline we need the curvatures and twist for all $s \in [0, L]$, and the pose of a distinguished cross-section. Let's take the distinguished cross-section to be at $s = s_0$. The evolution equation for the cross-section is

$$\partial_t \rho_0 = \rho_0 \psi_0 \tag{244}$$

where

$$\rho_0 = \rho\big|_{s=s_0}, \qquad \boldsymbol{\omega}_0 = \boldsymbol{\omega}_0|_{s=s_0}, \qquad \boldsymbol{u}_0 = \boldsymbol{u}|_{s=s_0}. \tag{245}$$

$$\psi_0 = \begin{pmatrix} \hat{\boldsymbol{\omega}}_0 & \boldsymbol{u}_0 \\ 0 & 0 \end{pmatrix}. \tag{246}$$

$$\rho_0(t+\mathrm{d}t) = \rho_0(t)\mathrm{O}\exp(\psi_0(t+\cdot);\mathrm{d}t) \tag{247}$$

$$= \rho_0(t)\exp\left(\frac{\mathrm{d}t}{2}\left(\psi_0(t+\mathrm{d}t)+\psi_0(t)\right)\right) + O(\mathrm{d}t^3) \tag{248}$$

2.6.1subsubsectionsection=0 . $s_0 = 0$

Evaluating (236),(237) at $s = 0$ results in

$$\boldsymbol{u}_0 = \frac{1}{c_\parallel}\left(\partial_s\lambda + f_a\right)\boldsymbol{e}_1 - \frac{B_\perp}{c_\perp}\partial_s^2\Omega_3\boldsymbol{e}_2 + \frac{B_\perp}{c_\perp}\partial_s^2\Omega_2\boldsymbol{e}_3 \tag{249}$$

$$\boldsymbol{\omega}_0 = \frac{1}{c_r}\left(B_\parallel\partial_s\Omega_1 + m_a\right)\boldsymbol{e}_1 - \frac{B_\perp}{c_\perp}\partial_s^3\Omega_2\boldsymbol{e}_2 - \frac{B_\perp}{c_\perp}\partial_s^3\Omega_3\boldsymbol{e}_3 \tag{250}$$

2.7subsectionsection=0 . Boundary and initial conditions

To solve the equations of motion (232-234) and integrate the frame/cernterline we need initial data for

- Cross-section s_0 pose — $\rho(s_0, 0)$
- Curvature vector at each $s \in [0, L]$ — $\boldsymbol{\Omega}(s, 0)$

and boundary conditions

- $\times 2$ for Ω_1
- $\times 4$ for $\boldsymbol{\Omega}_\perp$
- $\times 2$ for λ

Since we're interested in the case where the ends of the body are unconstrained, the boundary conditions are given by (102)

$$\boldsymbol{f}\big|_{s=0,L} = 0 \tag{251}$$

$$B\boldsymbol{\Omega}|_{s=0,L} = 0 \tag{252}$$

Using (235) to eliminate $\boldsymbol{f}_\perp$, we find

$$\Omega_1 = 0, \boldsymbol{\Omega}_\perp = 0, \partial_s \boldsymbol{\Omega}_\perp = 0, \lambda = 0. \tag{253}$$

at $s = 0$ and $s = L$.

2.8subsectionsection=0 . Nondimensional Equations 3D

In units with $L = 1, f_a^* = 1, c_\parallel = 1$ we have

$$\beta_\parallel = B_\parallel \tag{254}$$

$$\beta_\perp = B_\perp \tag{255}$$

$$\eta_\perp = c_\perp \tag{256}$$

$$\eta_r = c_r \tag{257}$$

$$U = 1 \tag{258}$$

$$
\begin{aligned}
\partial_t \Omega_1 + \left[\frac{\beta_\perp - \beta_\parallel}{\eta_\perp}|\boldsymbol{\Omega}_\perp|^2 - \frac{\beta_\parallel}{\eta_r}\right]\partial_s^2\Omega_1 &= G_1\left(\Omega_1, \partial_s\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \lambda, \partial_s m_a\right) \\
\partial_t \boldsymbol{\Omega}_\perp + \frac{\beta_\perp}{\eta_\perp}\partial_s^4\boldsymbol{\Omega}_\perp + \frac{\beta_\perp - \beta_\parallel}{\eta_\perp}\boldsymbol{e}_1 \times \boldsymbol{\Omega}_\perp \partial_s^3\Omega_1 &= G_\perp\left(\Omega_1, \ldots, \partial_s^2\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \lambda, \ldots, \partial_s^2\lambda, f_a, \partial_s f_a, m_a\right) \\
\partial_s^2\lambda - \frac{1}{\eta_\perp}|\boldsymbol{\Omega}_\perp|^2\lambda &= G_\lambda\left(\Omega_1, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^2\boldsymbol{\Omega}_\perp, \partial_s f_a\right) \\
\partial_t f_a &= \left[1 - f_a^2\right]u_1 + \frac{\sigma}{U}\partial_s^2 f_a
\end{aligned}
\tag{259}
$$

$$
\begin{aligned}
\Omega_1|_{s=0,L} &= 0 \\
\boldsymbol{\Omega}_\perp|_{s=0,L} &= 0 \\
\partial_s\boldsymbol{\Omega}_\perp|_{s=0,L} &= 0 \\
\lambda|_{s=0,L} &= 0 \\
\partial_s f_a\big|_{s=0,L} &= 0
\end{aligned}
\tag{260}
$$

3sectionsection=0 . 2D Nondimensional equations (no twist/active moment)

- Curvature/Angular velocity – $\boldsymbol{\Omega} = \kappa\boldsymbol{e}_z$, $\boldsymbol{\omega} = \omega\boldsymbol{e}_z$
- Stiffness – $\beta = \frac{B_\perp}{f_a^* L^3}$
- Drag – $\eta = \frac{c_\perp}{c_\parallel} \approx 1.585$
- Active force constant, Length – $f_a^* = 1$, $L = 1$

Curvature evolution

$$
\partial_t\kappa = -\frac{\beta}{\eta}\partial_s^4\kappa + \frac{\beta}{3}\partial_s^2\left(\kappa^3\right) + \frac{1}{\eta}\partial_s^2\left(\kappa\lambda\right) + \partial_s\left(\kappa\left[\partial_s\lambda + f\right]\right) \tag{261}
$$

$$\kappa|_{s=0,1} = 0, \quad \partial_s \kappa|_{s=0,1} = 0 \tag{262}$$

Tension equation

$$\partial_s^2 \lambda - \frac{\kappa^2}{\eta}\lambda = -\partial_s f - \frac{\beta}{2}\partial_s^2\left(\kappa^2\right) - \frac{\beta}{\eta}\kappa\partial_s^2\kappa \tag{263}$$

$$\lambda|_{s=0,1} = 0 \tag{264}$$

Active force evolution

$$\partial_t f = \sigma\partial_s^2 f + \left(1 - f^2\right)\left(\beta\kappa\partial_s\kappa + \partial_s\lambda + f\right) \tag{265}$$

$$\partial_s f\big|_{s=0,1} = 0 \tag{266}$$

Tangential Velocity

$$u_\parallel = \frac{\beta}{2}\partial_s\left(\kappa^2\right) + \partial_s\lambda + f \tag{267}$$

Normal Velocity

$$u_\perp = -\frac{\beta\partial_s^2\kappa}{\eta} + \frac{\kappa\lambda}{\eta} \tag{268}$$

Angular Velocity

$$\omega = \beta\kappa^2\partial_s\kappa + \kappa\partial_s\lambda + \kappa f + \frac{\kappa\partial_s\lambda}{\eta} + \frac{\lambda\partial_s\kappa}{\eta} - \frac{\beta\partial_s^3\kappa}{\eta} \tag{269}$$

Some conserved stuff

$$\int_0^1 u_\parallel - f \,\mathrm{d}s = 0 \tag{270}$$

$$\int_0^1 u_\perp - \frac{1}{\eta}\kappa\lambda \,\mathrm{d}s = 0 \tag{271}$$

3.1subsectionsection=0 . Steady state

Note: if $f = -1\Theta(s-1/2)+1$ you need to match bcs at $s = 1/2$

$$-\frac{\beta}{\eta}\partial_s^4\kappa + \frac{\beta}{3}\partial_s^2\left(\kappa^3\right) + \frac{1}{\eta}\partial_s^2(\kappa\lambda) + \partial_s\left(\kappa\left[\partial_s\lambda + f\right]\right) = 0 \tag{272}$$

$$\partial_s^4\kappa - \frac{\eta}{3}\partial_s^2\left(\kappa^3\right) - \frac{1}{\beta}\partial_s^2(\kappa\lambda) - \frac{\eta}{\beta}\partial_s\left(\kappa\left[\partial_s\lambda + f\right]\right) = 0 \tag{273}$$

$$\partial_s^3\kappa - \frac{\eta}{3}\partial_s\left(\kappa^3\right) - \frac{1}{\beta}\partial_s(\lambda\kappa) - \frac{\eta}{\beta}\left[\partial_s\lambda + f\right]\kappa = a \tag{274}$$

For some constante a. Evaluate at $s = 0,1$ and apply boundary conditions to get

$$\partial_s^3\kappa_{s=0,1} = a \tag{275}$$

3.2subsectionsection=0 . Tension integral equation

The tension differential equation (263) is equivalent to the Fredholm equation

$$\lambda(s) = -\frac{\beta}{2}\kappa^2(s) - \int_0^1 G(s,z)f'(z)\,\mathrm{d}z - \frac{\beta}{2\eta}\kappa^2(s) + \frac{\beta}{\eta}\int_0^1 G(s,z)[\kappa'(z)]^2\,\mathrm{d}z + \frac{1}{\eta}\int_0^1 G(s,z)\kappa^2(z)\lambda(z)\,\mathrm{d}z \tag{276}$$

where G is the Dirichlet Green function for the Laplace equation,

$$G(s,s') = \begin{cases} (s'-1)s, & \text{for } s < s' \\ s'(s-1), & \text{for } s > s' \end{cases} \tag{277}$$

$$= ss' - s\Theta(s'-s) - s'\Theta(s-s'). \tag{278}$$

To simplify notation, define the following:

$$\varepsilon = \frac{1}{\eta} \tag{279}$$

$$g_0(s) = -\frac{\beta}{2}\kappa^2(s) - \int_0^1 G(s,z) f'(z)\,\mathrm{d}z \tag{280}$$

$$g_1(s) = -\frac{\beta}{2}\kappa^2(s) + \beta\int_0^1 G(s,z)[\kappa'(z)]^2\,\mathrm{d}z \tag{281}$$

$$K(x,y) = G(x,y)\kappa^2(y). \tag{282}$$

Equation (276) is then

$$|\lambda\rangle = \left|g_0\right\rangle + \varepsilon\left|g_1\right\rangle + \varepsilon\hat{K}|\lambda\rangle \tag{283}$$

has solution

$$|\lambda\rangle = \left|g_0\right\rangle + \sum_{j=1} \varepsilon^j \hat{K}^{j-1}\left[\hat{K}\left|g_0\right\rangle + \left|g_1\right\rangle\right]. \tag{284}$$

3.2.1subsubsectionsection=0 . First few terms

$$\lambda_0 = -\frac{\beta}{2}\kappa^2(s) - \int_0^1 G(s,z) f'(z)\,\mathrm{d}z \tag{285}$$

$$\lambda_1 = -\frac{\beta}{2}\kappa^2(s) + \beta\int_0^1 G(s,z)[\kappa'(z)]^2\,\mathrm{d}z - \frac{\beta}{2}\int_0^1 G(s,z)\kappa^4(z)\,\mathrm{d}z - \int_0^1\int_0^1 G(s,y)\kappa^2(y)G(y,z)f'(z)\,\mathrm{d}z\,\mathrm{d}y \tag{286}$$

3.2.2subsubsectionsection=0 . $f(s) = 1 - 2\Theta(s-1/2)$

If $f = -2\Theta(s-1/2)+1$ then $\partial_s = -2\delta(s-1/2)$ and

$$-\int_0^1 \mathrm{d}z\, G(s,z)\partial_z f(z) = 2G(s,1/2) \tag{287}$$

$$= s - 2s\Theta(1/2-s) - \Theta(s-1/2) \tag{288}$$

$$g_0(s) = -\frac{\beta}{2}\kappa^2(s) + s - 2s\Theta(1/2-s) - \Theta(s-1/2) \tag{289}$$

3.3subsectionsection=0 . Constant curvature

$$f = -2\Theta(s-1/2)+1 \tag{290}$$

$$\partial_s f = -2\delta(s-1/2) \tag{291}$$

$$\partial_s^2\lambda - \frac{\kappa^2}{\eta}\lambda = 2\delta(s-1/2) \tag{292}$$

$$\partial_s\lambda(1/2^+) - \partial_s\lambda(1/2^-) = 2 \tag{293}$$

$$\lambda(0) = \lambda(1) = 0 \tag{294}$$

$$\lambda = \begin{cases} \frac{-\sqrt{\eta}}{\kappa \cosh\left(\frac{\kappa}{2\sqrt{\eta}}\right)} \sinh\left(\frac{\kappa}{\sqrt{\eta}} s\right), & \text{for } s < 1/2 \\ \frac{\sqrt{\eta}}{\kappa \cosh\left(\frac{\kappa}{2\sqrt{\eta}}\right)} \sinh\left(\frac{\kappa}{\sqrt{\eta}} (s-1)\right), & \text{for } s > 1/2 \end{cases} \tag{295}$$

$$= \begin{cases} -s + O(\kappa^2), & \text{for } s < 1/2 \\ s - 1 + O(\kappa^2), & \text{for } s > 1/2 \end{cases} \tag{296}$$

No swimming without $O(\kappa^2)$ terms when κ =constant?

$$\partial_s \lambda = 2\Theta(s - 1/2) - 1 + O(\kappa^2) \tag{297}$$

$$u_\parallel = \partial_s \lambda + f = O(\kappa^2) \tag{298}$$

$$u_\perp = \frac{\kappa \lambda}{\eta} = \begin{cases} -\frac{\kappa s}{\eta} + O(\kappa^3), & \text{for } s < 1/2 \\ \frac{\kappa(s-1)}{\eta} + O(\kappa^3), & \text{for } s > 1/2 \end{cases} \tag{299}$$

3.4subsectionsection=0 . Linear stability (2D)

Throw out all $O(\kappa^2)$ terms and let $\overline{f} = \int_0^1 f \, \mathrm{d}s$ be the average active force. Then, the tension satisfies,

$$\lambda = \overline{f} s - \int_0^s f \, \mathrm{d}s = -\int_0^s (f - \overline{f}) \, \mathrm{d}s \tag{300}$$

$$\partial_s \lambda + f = \overline{f}, \tag{301}$$

and the curvature is governed by

$$\partial_t \kappa = -\frac{\beta}{\eta} \partial_s^4 \kappa - \frac{1}{\eta} \partial_s^2 \left(\kappa \int_0^s f - \overline{f} \, \mathrm{d}s \right) + \overline{f} \partial_s \kappa, \tag{302}$$

Proof: Integrate $\partial_s^2 \lambda = -\partial_s f$ and use $\lambda(0) = \lambda(1) = 0$ to get $\lambda = \overline{f} s - \int_0^s f \, \mathrm{d}s$ and $\partial_s \lambda + f = \overline{f}$. Plug the tension expression into the linearized curvature equation

$$\partial_t \kappa = -\frac{\beta}{\eta} \partial_s^4 \kappa + \frac{1}{\eta} \partial_s^2 (\kappa \lambda) + \partial_s \left(\kappa \left[\partial_s \lambda + f \right] \right) \tag{303}$$

$$= -\frac{\beta}{\eta} \partial_s^4 \kappa + \frac{1}{\eta} \partial_s^2 \left(\left[\overline{f} s - \int_0^s f \, \mathrm{d}s \right] \kappa \right) + \overline{f} \partial_s \kappa \tag{304}$$

□

3.4.1subsubsectionsection=0 . $\overline{f} = 0$

When the average active force vanishes, $\overline{f} = 0$,

$$\lambda = -\int_0^s f \, \mathrm{d}s \tag{305}$$

$$\partial_t \kappa = -\frac{\beta}{\eta} \partial_s^4 \kappa - \frac{1}{\eta} \partial_s^2 \left(\kappa \int_0^s f \, \mathrm{d}s \right) \tag{306}$$

$$\mathscr{L}[\kappa_n] = \chi_n \kappa_n = -\frac{\beta}{\eta} \partial_s^4 \kappa_n - \frac{1}{\eta} \partial_s^2 \left(\kappa_n \int_0^s f \, \mathrm{d}s \right) \tag{307}$$

3.4.2subsubsectionsection=0 . Piecewise constant active force

$$f = \begin{cases} 1 & \text{, when } s < 1/2 \\ -1 & \text{, when } s > 1/2 \end{cases} \tag{308}$$

$$\lambda = \begin{cases} -s & \text{, when } s < 1/2 \\ s - 1 & \text{, when } s > 1/2 \end{cases} \tag{309}$$

$$\partial_t \kappa = \begin{cases} -\frac{\beta}{\eta}\partial_s^2\left(\partial_s^2 + \frac{s}{\beta}\right)\kappa & \text{, when } s < 1/2 \\ -\frac{\beta}{\eta}\partial_s^2\left(\partial_s^2 + \frac{1-s}{\beta}\right)\kappa & \text{, when } s > 1/2 \end{cases} \tag{310}$$

3.4.3subsubsectionsection=0 . Regularized piecewise constant active force

$$f = -\tanh(a(s-1/2)) \tag{311}$$

$$f \to -2\Theta(s-1/2) + 1 \text{ as } a \to \infty \tag{312}$$

Tension

$$\partial_s^2 \lambda = -\partial_s f = a\,\mathrm{sech}^2(a(s-1/2)) \tag{313}$$

$$\partial_s \lambda = \tanh(a(s-1/2)) + b \tag{314}$$

$$\lambda = \ln(\cosh(a(s-1/2)))/a + bs + c \tag{315}$$

$$\lambda|_{s=0,1} = 0 \Rightarrow b = 0, c = -\ln(\cosh(a/2))/a \tag{316}$$

$$\lambda = \frac{1}{a}\ln\left(\frac{\cosh(a(s-1/2))}{\cosh(a/2)}\right) \tag{317}$$

since $\partial_s \lambda + f = 0$, the curvature equation reduces to

$$\partial_t \kappa = -\frac{\beta}{\eta}\partial_s^4 \kappa + \frac{1}{\eta}\partial_s^2 \left(\frac{1}{a}\ln\left(\frac{\cosh(a(s-1/2))}{\cosh(a/2)}\right)\kappa\right) \tag{318}$$

3.5subsectionsection=0 . Linear reduced order model

•

4sectionsection=0 . 3D Sympy expressions

Twist

$$\begin{aligned}
&-\frac{M\partial_s f}{\eta_r} + \frac{\Omega_0^3\Omega_1^2\beta_\parallel}{\eta} - \frac{\Omega_0^3\Omega_1^2\beta_\perp}{\eta} + \frac{\Omega_0^3\Omega_2^2\beta_\parallel}{\eta} - \frac{\Omega_0^3\Omega_2^2\beta_\perp}{\eta} \\
&+\frac{2\Omega_0^2\Omega_1\beta_\parallel\partial_s\Omega_2}{\eta} - \frac{3\Omega_0^2\Omega_1\beta_\perp\partial_s\Omega_2}{\eta} - \frac{2\Omega_0^2\Omega_2\beta_\parallel\partial_s\Omega_1}{\eta} + \frac{3\Omega_0^2\Omega_2\beta_\perp\partial_s\Omega_1}{\eta} - \frac{\Omega_0\Omega_1^2\lambda}{\eta} \\
&-\frac{\Omega_0\Omega_1\beta_\parallel\partial_s^2\Omega_1}{\eta} + \frac{3\Omega_0\Omega_1\beta_\perp\partial_s^2\Omega_1}{\eta} - \frac{\Omega_0\Omega_2^2\lambda}{\eta} - \frac{\Omega_0\Omega_2\beta_\parallel\partial_s^2\Omega_2}{\eta} + \frac{3\Omega_0\Omega_2\beta_\perp\partial_s^2\Omega_2}{\eta} \\
&-\frac{\Omega_1^2\beta_\parallel\partial_s^2\Omega_0}{\eta} + \frac{\Omega_1^2\beta_\perp\partial_s^2\Omega_0}{\eta} - \frac{2\Omega_1\beta_\parallel\partial_s\Omega_0\partial_s\Omega_1}{\eta} + \frac{3\Omega_1\beta_\perp\partial_s\Omega_0\partial_s\Omega_1}{\eta} + \frac{\Omega_1\beta_\perp\partial_s^3\Omega_2}{\eta} \\
&-\frac{\Omega_1\lambda\partial_s\Omega_2}{\eta} - \frac{\Omega_2^2\beta_\parallel\partial_s^2\Omega_0}{\eta} + \frac{\Omega_2^2\beta_\perp\partial_s^2\Omega_0}{\eta} - \frac{2\Omega_2\beta_\parallel\partial_s\Omega_0\partial_s\Omega_2}{\eta} + \frac{3\Omega_2\beta_\perp\partial_s\Omega_0\partial_s\Omega_2}{\eta} \\
&-\frac{\Omega_2\beta_\perp\partial_s^3\Omega_1}{\eta} + \frac{\Omega_2\lambda\partial_s\Omega_1}{\eta} - \frac{\beta_\parallel\partial_s^2\Omega_0}{\eta_r} + \partial_t\Omega_0 = 0
\end{aligned} \tag{319}$$

Curvature1

$$
\begin{aligned}
&-\frac{M\Omega_2 f}{\eta_r}-\frac{\Omega_0^4\Omega_1\beta_\parallel}{\eta}+\frac{\Omega_0^4\Omega_1\beta_\perp}{\eta}-\frac{3\Omega_0^3\beta_\parallel\partial_s\Omega_2}{\eta}+\frac{4\Omega_0^3\beta_\perp\partial_s\Omega_2}{\eta}\\
&+\frac{\Omega_0^2\Omega_1\lambda}{\eta}-\frac{6\Omega_0^2\Omega_2\beta_\parallel\partial_s\Omega_0}{\eta}+\frac{6\Omega_0^2\Omega_2\beta_\perp\partial_s\Omega_0}{\eta}+\frac{3\Omega_0^2\beta_\parallel\partial_s^2\Omega_1}{\eta}-\frac{6\Omega_0^2\beta_\perp\partial_s^2\Omega_1}{\eta}\\
&+\Omega_0\Omega_1\Omega_2\beta_\perp\partial_s\Omega_1+\frac{4\Omega_0\Omega_1\beta_\parallel\partial_s^2\Omega_0}{\eta}-\frac{4\Omega_0\Omega_1\beta_\perp\partial_s^2\Omega_0}{\eta}+\Omega_0\Omega_2^2\beta_\perp\partial_s\Omega_2+\Omega_0\Omega_2\partial_s\lambda\\
&+\Omega_0\Omega_2 f+\frac{2\Omega_0\Omega_2\partial_s\lambda}{\eta}+\frac{9\Omega_0\beta_\parallel\partial_s\Omega_0\partial_s\Omega_1}{\eta}+\frac{\Omega_0\beta_\parallel\partial_s^3\Omega_2}{\eta}-\frac{12\Omega_0\beta_\perp\partial_s\Omega_0\partial_s\Omega_1}{\eta}\\
&-\frac{4\Omega_0\beta_\perp\partial_s^3\Omega_2}{\eta}+\frac{2\Omega_0\lambda\partial_s\Omega_2}{\eta}-\Omega_1^2\beta_\perp\partial_s^2\Omega_1-\Omega_1\Omega_2\beta_\perp\partial_s^2\Omega_2+\frac{3\Omega_1\beta_\parallel\partial_s\Omega_0^2}{\eta}\\
&-2\Omega_1\beta_\perp\partial_s\Omega_1^2-\Omega_1\beta_\perp\partial_s\Omega_2^2-\frac{3\Omega_1\beta_\perp\partial_s\Omega_0^2}{\eta}-\Omega_1\partial_s^2\lambda-\Omega_1\partial_s f\\
&-\frac{\Omega_1\partial_s^2\lambda}{\eta}-\frac{\Omega_2\beta_\parallel\partial_s\Omega_0}{\eta_r}+\frac{\Omega_2\beta_\parallel\partial_s^3\Omega_0}{\eta}-\Omega_2\beta_\perp\partial_s\Omega_1\partial_s\Omega_2-\frac{\Omega_2\beta_\perp\partial_s^3\Omega_0}{\eta}\\
&+\frac{\Omega_2\lambda\partial_s\Omega_0}{\eta}+\frac{3\beta_\parallel\partial_s\Omega_0\partial_s^2\Omega_2}{\eta}+\frac{3\beta_\parallel\partial_s\Omega_2\partial_s^2\Omega_0}{\eta}-\frac{6\beta_\perp\partial_s\Omega_0\partial_s^2\Omega_2}{\eta}-\frac{4\beta_\perp\partial_s\Omega_2\partial_s^2\Omega_0}{\eta}\\
&+\frac{\beta_\perp\partial_s^4\Omega_1}{\eta}-\partial_s\Omega_1\partial_s\lambda-\partial_s\Omega_1 f+\partial_t\Omega_1-\frac{\lambda\partial_s^2\Omega_1}{\eta}\\
&-\frac{2\partial_s\Omega_1\partial_s\lambda}{\eta}=0
\end{aligned}
\tag{320}
$$

Curvature2

$$\frac{M\Omega_1 f}{\eta_r} - \frac{\Omega_0^4\Omega_2\beta_\parallel}{\eta} + \frac{\Omega_0^4\Omega_2\beta_\perp}{\eta} + \frac{3\Omega_0^3\beta_\parallel\partial_s\Omega_1}{\eta} - \frac{4\Omega_0^3\beta_\perp\partial_s\Omega_1}{\eta}$$
$$+ \frac{6\Omega_0^2\Omega_1\beta_\parallel\partial_s\Omega_0}{\eta} - \frac{6\Omega_0^2\Omega_1\beta_\perp\partial_s\Omega_0}{\eta} + \frac{\Omega_0^2\Omega_2\lambda}{\eta} + \frac{3\Omega_0^2\beta_\parallel\partial_s^2\Omega_2}{\eta} - \frac{6\Omega_0^2\beta_\perp\partial_s^2\Omega_2}{\eta}$$
$$- \Omega_0\Omega_1^2\beta_\perp\partial_s\Omega_1 - \Omega_0\Omega_1\Omega_2\beta_\perp\partial_s\Omega_2 - \Omega_0\Omega_1\partial_s\lambda - \Omega_0\Omega_1 f - \frac{2\Omega_0\Omega_1\partial_s\lambda}{\eta}$$
$$+ \frac{4\Omega_0\Omega_2\beta_\parallel\partial_s^2\Omega_0}{\eta} - \frac{4\Omega_0\Omega_2\beta_\perp\partial_s^2\Omega_0}{\eta} + \frac{9\Omega_0\beta_\parallel\partial_s\Omega_0\partial_s\Omega_2}{\eta} - \frac{\Omega_0\beta_\parallel\partial_s^3\Omega_1}{\eta} - \frac{12\Omega_0\beta_\perp\partial_s\Omega_0\partial_s\Omega_2}{\eta}$$
$$+ \frac{4\Omega_0\beta_\perp\partial_s^3\Omega_1}{\eta} - \frac{2\Omega_0\lambda\partial_s\Omega_1}{\eta} - \Omega_1\Omega_2\beta_\perp\partial_s^2\Omega_1 + \frac{\Omega_1\beta_\parallel\partial_s\Omega_0}{\eta_r} - \frac{\Omega_1\beta_\parallel\partial_s^3\Omega_0}{\eta}$$
$$- \Omega_1\beta_\perp\partial_s\Omega_1\partial_s\Omega_2 + \frac{\Omega_1\beta_\perp\partial_s^3\Omega_0}{\eta} - \frac{\Omega_1\lambda\partial_s\Omega_0}{\eta} - \Omega_2^2\beta_\perp\partial_s^2\Omega_2 + \frac{3\Omega_2\beta_\parallel\partial_s\Omega_0^2}{\eta}$$
$$- \Omega_2\beta_\perp\partial_s\Omega_1^2 - 2\Omega_2\beta_\perp\partial_s\Omega_2^2 - \frac{3\Omega_2\beta_\perp\partial_s\Omega_0^2}{\eta} - \Omega_2\partial_s^2\lambda - \Omega_2\partial_s f$$
$$- \frac{\Omega_2\partial_s^2\lambda}{\eta} - \frac{3\beta_\parallel\partial_s\Omega_0\partial_s^2\Omega_1}{\eta} - \frac{3\beta_\parallel\partial_s\Omega_1\partial_s^2\Omega_0}{\eta} + \frac{6\beta_\perp\partial_s\Omega_0\partial_s^2\Omega_1}{\eta} + \frac{4\beta_\perp\partial_s\Omega_1\partial_s^2\Omega_0}{\eta}$$
$$+ \frac{\beta_\perp\partial_s^4\Omega_2}{\eta} - \partial_s\Omega_2\partial_s\lambda - \partial_s\Omega_2 f + \partial_t\Omega_2 - \frac{\lambda\partial_s^2\Omega_2}{\eta}$$
$$- \frac{2\partial_s\Omega_2\partial_s\lambda}{\eta} = 0 \tag{321}$$

Active force

$$-\partial_s^2 f\sigma + \partial_t f - \left(1 - f^2\right)\left(\Omega_1\beta_\perp\partial_s\Omega_1 + \Omega_2\beta_\perp\partial_s\Omega_2 + \partial_s\lambda + f\right) = 0 \tag{322}$$

Tension

$$\begin{aligned}
&\frac{\Omega_0^2\Omega_1^2\beta_\parallel}{\eta} - \frac{\Omega_0^2\Omega_1^2\beta_\perp}{\eta} + \frac{\Omega_0^2\Omega_2^2\beta_\parallel}{\eta} - \frac{\Omega_0^2\Omega_2^2\beta_\perp}{\eta} + \frac{\Omega_0\Omega_1\beta_\parallel\partial_s\Omega_2}{\eta} \\
&- \frac{2\Omega_0\Omega_1\beta_\perp\partial_s\Omega_2}{\eta} - \frac{\Omega_0\Omega_2\beta_\parallel\partial_s\Omega_1}{\eta} + \frac{2\Omega_0\Omega_2\beta_\perp\partial_s\Omega_1}{\eta} - \frac{\Omega_1^2\lambda}{\eta} + \Omega_1\beta_\perp\partial_s^2\Omega_1 \\
&+ \frac{\Omega_1\beta_\perp\partial_s^2\Omega_1}{\eta} - \frac{\Omega_2^2\lambda}{\eta} + \Omega_2\beta_\perp\partial_s^2\Omega_2 + \frac{\Omega_2\beta_\perp\partial_s^2\Omega_2}{\eta} + \beta_\perp\partial_s\Omega_1^2 \\
&+ \beta_\perp\partial_s\Omega_2^2 + \partial_s^2\lambda + \partial_s f = 0
\end{aligned} \tag{323}$$

Angular velocity0

$$\omega_0 = \frac{Mf}{\eta_r} + \frac{\beta_\parallel\partial_s\Omega_0}{\eta_r} \tag{324}$$

Angular velocity1

$$\begin{aligned}
\omega_1 =& \frac{\Omega_0^3\Omega_2\beta_\parallel}{\eta} - \frac{\Omega_0^3\Omega_2\beta_\perp}{\eta} - \frac{2\Omega_0^2\beta_\parallel\partial_s\Omega_1}{\eta} + \frac{3\Omega_0^2\beta_\perp\partial_s\Omega_1}{\eta} - \frac{3\Omega_0\Omega_1\beta_\parallel\partial_s\Omega_0}{\eta} \\
&+ \frac{3\Omega_0\Omega_1\beta_\perp\partial_s\Omega_0}{\eta} - \frac{\Omega_0\Omega_2\lambda}{\eta} - \frac{\Omega_0\beta_\parallel\partial_s^2\Omega_2}{\eta} + \frac{3\Omega_0\beta_\perp\partial_s^2\Omega_2}{\eta} + \Omega_1^2\beta_\perp\partial_s\Omega_1 \\
&+ \Omega_1\Omega_2\beta_\perp\partial_s\Omega_2 + \Omega_1\partial_s\lambda + \Omega_1 f + \frac{\Omega_1\partial_s\lambda}{\eta} - \frac{\Omega_2\beta_\parallel\partial_s^2\Omega_0}{\eta} \\
&+ \frac{\Omega_2\beta_\perp\partial_s^2\Omega_0}{\eta} - \frac{2\beta_\parallel\partial_s\Omega_0\partial_s\Omega_2}{\eta} + \frac{3\beta_\perp\partial_s\Omega_0\partial_s\Omega_2}{\eta} - \frac{\beta_\perp\partial_s^3\Omega_1}{\eta} + \frac{\lambda\partial_s\Omega_1}{\eta}
\end{aligned} \tag{325}$$

Angular velocity2

$$\begin{aligned}
\omega_2 =& -\frac{\Omega_0^3\Omega_1\beta_\parallel}{\eta} + \frac{\Omega_0^3\Omega_1\beta_\perp}{\eta} - \frac{2\Omega_0^2\beta_\parallel\partial_s\Omega_2}{\eta} + \frac{3\Omega_0^2\beta_\perp\partial_s\Omega_2}{\eta} + \frac{\Omega_0\Omega_1\lambda}{\eta} \\
&- \frac{3\Omega_0\Omega_2\beta_\parallel\partial_s\Omega_0}{\eta} + \frac{3\Omega_0\Omega_2\beta_\perp\partial_s\Omega_0}{\eta} + \frac{\Omega_0\beta_\parallel\partial_s^2\Omega_1}{\eta} - \frac{3\Omega_0\beta_\perp\partial_s^2\Omega_1}{\eta} + \Omega_1\Omega_2\beta_\perp\partial_s\Omega_1 \\
&+ \frac{\Omega_1\beta_\parallel\partial_s^2\Omega_0}{\eta} - \frac{\Omega_1\beta_\perp\partial_s^2\Omega_0}{\eta} + \Omega_2^2\beta_\perp\partial_s\Omega_2 + \Omega_2\partial_s\lambda + \Omega_2 f \\
&+ \frac{\Omega_2\partial_s\lambda}{\eta} + \frac{2\beta_\parallel\partial_s\Omega_0\partial_s\Omega_1}{\eta} - \frac{3\beta_\perp\partial_s\Omega_0\partial_s\Omega_1}{\eta} - \frac{\beta_\perp\partial_s^3\Omega_2}{\eta} + \frac{\lambda\partial_s\Omega_2}{\eta}
\end{aligned} \tag{326}$$

Linear velocity0

$$u_0 = \Omega_1 \beta_\perp \partial_s \Omega_1 + \Omega_2 \beta_\perp \partial_s \Omega_2 + \partial_s \lambda + f \tag{327}$$

Linear velocity1

$$\begin{aligned} u_1 = &- \frac{\Omega_0^2 \Omega_2 \beta_\parallel}{\eta} + \frac{\Omega_0^2 \Omega_2 \beta_\perp}{\eta} + \frac{\Omega_0 \beta_\parallel \partial_s \Omega_1}{\eta} - \frac{2\Omega_0 \beta_\perp \partial_s \Omega_1}{\eta} + \frac{\Omega_1 \beta_\parallel \partial_s \Omega_0}{\eta} \\ &- \frac{\Omega_1 \beta_\perp \partial_s \Omega_0}{\eta} + \frac{\Omega_2 \lambda}{\eta} - \frac{\beta_\perp \partial_s^2 \Omega_2}{\eta} \end{aligned} \tag{328}$$

Linear velocity2

$$\begin{aligned} u_2 = &\frac{\Omega_0^2 \Omega_1 \beta_\parallel}{\eta} - \frac{\Omega_0^2 \Omega_1 \beta_\perp}{\eta} + \frac{\Omega_0 \beta_\parallel \partial_s \Omega_2}{\eta} - \frac{2\Omega_0 \beta_\perp \partial_s \Omega_2}{\eta} - \frac{\Omega_1 \lambda}{\eta} \\ &+ \frac{\Omega_2 \beta_\parallel \partial_s \Omega_0}{\eta} - \frac{\Omega_2 \beta_\perp \partial_s \Omega_0}{\eta} + \frac{\beta_\perp \partial_s^2 \Omega_1}{\eta} \end{aligned} \tag{329}$$

4.0.1subsubsectionsection=0 . 3D Linearized Equations

$$\partial_t \Omega_0 - \frac{\beta_\parallel}{\eta_r} \partial_s^2 \Omega_0 - \frac{M}{\eta_r} \partial_s f = 0 \tag{330}$$

$$\partial_t \Omega_1 + \frac{\beta_\perp}{\eta_\perp} \partial_s^4 \Omega_1 = 0 \tag{331}$$

$$\partial_t \Omega_2 + \frac{\beta_\perp}{\eta_\perp} \partial_s^4 \Omega_2 = 0 \tag{332}$$

$$\partial_t f - \sigma \partial_s^2 f - f - \partial_s \lambda = 0 \tag{333}$$

$$\partial_s^2 \lambda + \partial_s f = 0 \tag{334}$$

5sectionsection=0 . Trajectories

...

6sectionsection=0 . 3D Numerics

The equations of motion (259) can be written in the form

$$\begin{aligned}
\partial_t\Omega_0 - \dot{\Omega}_0\left(\Omega_0, \partial_s\Omega_0, \partial_s^2\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^3\boldsymbol{\Omega}_\perp, \partial_s f, \lambda\right) &= 0 \\
\partial_t\Omega_1 - \dot{\Omega}_1\left(\Omega_0, \ldots, \partial_s^3\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^4\boldsymbol{\Omega}_\perp, f, \partial_s f, \lambda, \ldots, \partial_s^2\lambda\right) &= 0 \\
\partial_t\Omega_2 - \dot{\Omega}_2\left(\Omega_0, \ldots, \partial_s^3\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^4\boldsymbol{\Omega}_\perp, f, \partial_s f, \lambda, \ldots, \partial_s^2\lambda\right) &= 0 \\
\partial_t f - \dot{f}(\boldsymbol{\Omega}_\perp, \partial_s\boldsymbol{\Omega}_\perp, f, \partial_s^2 f, \partial_s\lambda) &= 0 \\
\Lambda\left(\Omega_0, \boldsymbol{\Omega}_\perp, \ldots, \partial_s^2\boldsymbol{\Omega}_\perp, \partial_s f, \lambda, \partial_s^2\lambda\right) &= 0
\end{aligned} \tag{335}$$

$$\begin{aligned}
\Omega_0|_{s=0,L}, \Omega_1|_{s=0,L}, \partial_s\Omega_1|_{s=0,L}, \Omega_2|_{s=0,L} &= 0 \\
\partial_s\Omega_2|_{s=0,L}, \partial_s f\big|_{s=0,L}, \lambda|_{s=0,L} &= 0
\end{aligned} \tag{336}$$

To simplify some calculations, we're just going to think of $\dot{\Omega}_0, \dot{\Omega}_1, \dot{\Omega}_2, \dot{f}, \Lambda$ as functions of all variables and their first four derivatives. That is, we treat them as maps

$$\begin{aligned}
\dot{\Omega}_0 &: \mathbb{R}^{25} \to \mathbb{R} \\
\dot{\Omega}_1 &: \mathbb{R}^{25} \to \mathbb{R} \\
\dot{\Omega}_2 &: \mathbb{R}^{25} \to \mathbb{R} \\
\dot{f} &: \mathbb{R}^{25} \to \mathbb{R} \\
\Lambda &: \mathbb{R}^{25} \to \mathbb{R}
\end{aligned} \tag{337}$$

More notation

$$X^p := \begin{cases} \Omega_0, & p = 0 \\ \Omega_1, & p = 1 \\ \Omega_2, & p = 2 \\ f, & p = 3 \\ \lambda, & p = 4 \end{cases} \tag{338}$$

Denote canonical basis vectors for $\mathbb{R}^5$ by $\left|p\right\rangle$ and the vector samples at s by

$$X := \sum_{p\in\mathbb{Z}_5} X^p \left|p\right\rangle. \tag{339}$$

For each $p \in \mathbb{Z}^5$, define

$$G^p : \mathbb{R}^{25} \to \mathbb{R} \tag{340}$$

$$\tag{341}$$

$$\begin{aligned}
G^0(y_q^p) &= -\dot{\Omega}_0(y_q^p) \\
G^1(y_q^p) &= -\dot{\Omega}_1(y_q^p) \\
G^2(y_q^p) &= -\dot{\Omega}_2(y_q^p) \\
G^3(y_q^p) &= -\dot{f}(y_q^p) \\
G^4(y_q^p) &= \Lambda(y_q^p)
\end{aligned} \tag{342}$$

where y_q^p is shorthand for

$$(y_0^0, y_0^1, \ldots, y_0^4, \ldots y_4^4, \ldots y_4^4) \in \mathbb{R}^{25} \tag{343}$$

The system (335) can then be written

$$(1-\delta_4^p)\partial_t X^p + G^p(X, \ldots \partial_s^4 X) = 0, \text{ for } p \in \mathbb{Z}_5 \tag{344}$$

6.1subsectionsection=0 . BDF2 with Newton

This section has a bunch of calculations you need to solve the discretization of (344) using Newton's method.

- Arclength index – $s \in \{-2, -1, 0, \ldots, N-1, N, N+1\}$
- Python index $j \in \mathbb{Z}_{N+4}$, $j \equiv s (\bmod N+4)$

- Variable index – $p \in \mathbb{Z}_5$

- Derivative index – $q \in \mathbb{Z}_6$

Let $\mathscr{H}$ be a $(5N+20)$-dimensional real Hilbert space with orthonormal basis,

$$\left|sp\right\rangle = |s\rangle \otimes \left|p\right\rangle \text{ for } -2 \leq s \leq N+1 \text{ and } p \in \mathbb{Z}_5. \tag{345}$$

Write state vectors as

$$|X\rangle = \sum_{s=-2}^{N+1} \sum_{p \in \mathbb{Z}_5} X^{sp} \left|sp\right\rangle \tag{346}$$

where X^{sp} represents the sample of the p-th variable at grid point s. When neccessary, we'll also add an additional time superscript,

$$X^{spt}, X^{sp,t-1}, \text{etc}\ldots \tag{347}$$

Define the 2nd order accurate difference operators

$$D^{(q)} = \sum_{|s-s'| \leq 2} \sum_{p \in \mathbb{Z}_5} D^{(q)}_{ss'} \left|sp\right\rangle \left\langle s'p\right| \tag{348}$$

$$D^{(q)}_{ss'} \delta_{pp'} = \left\langle sp\right| D^{(q)} \left|s'p'\right\rangle \tag{349}$$

$$D^{(0)}_{ss'} = \delta_{ss'} \tag{350}$$

$$D^{(1)}_{ss'} = -\frac{1}{2\,\mathrm{d}s}\delta_{s-1,s'} + \frac{1}{2\,\mathrm{d}s}\delta_{s+1,s'} \tag{351}$$

$$D^{(2)}_{ss'} = \frac{1}{\mathrm{d}s^2}\delta_{s-1,s'} - \frac{2}{\mathrm{d}s^2}\delta_{s,s'} + \frac{1}{\mathrm{d}s^2}\delta_{s+1,s'} \tag{352}$$

$$D^{(3)}_{ss'} = -\frac{1}{2\,\mathrm{d}s^3}\delta_{s-2,s'} + \frac{1}{\mathrm{d}s^3}\delta_{s-1,s'} - \frac{1}{\mathrm{d}s^3}\delta_{s+1,s'} + \frac{1}{2\,\mathrm{d}s^3}\delta_{s+2,s'} \tag{353}$$

$$D^{(4)}_{ss'} = \frac{1}{\mathrm{d}s^4}\delta_{s-2,s'} - \frac{4}{\mathrm{d}s^4}\delta_{s-1,s'} + \frac{6}{\mathrm{d}s^4}\delta_{s,s'} - \frac{4}{\mathrm{d}s^4}\delta_{s+1,s'} + \frac{1}{\mathrm{d}s^4}\delta_{s+2,s'} \tag{354}$$

$$D_{tt'} = \frac{1}{2\,\mathrm{d}t}\delta_{t-2,t'} - \frac{2}{\mathrm{d}t}\delta_{t-1,t'} + \frac{3}{2\,\mathrm{d}t}\delta_{t,t'} \tag{355}$$

and write finite differences of variables as

$$X^{sp}_q := \begin{cases} \sum\limits_{|s'-s|\le 2} D^{(q)}_{ss'} X^{s'p}, & \text{for } q \in \mathbb{Z}_5 \\ \frac{1}{2\mathrm{d}t} X^{s,t-2,p} - \frac{2}{\mathrm{d}t} X^{s,t-1,p} + \frac{3}{2\mathrm{d}t} X^{s,t,p}, & \text{for } q = 5. \end{cases} \tag{356}$$

For $q \in \mathbb{Z}_5$

$$X^{sp}_q = X^{sp}_q(X^{s-2,p}, X^{s-1,p}, X^{s,p}, X^{s+1,p}, X^{s+2,p}). \tag{357}$$

Write $G^p(\ldots X^{sp'}_q \ldots)$ as

$$G^{sp} := G^p(\ldots X^{sp'}_q \ldots). \tag{358}$$

$$G^{sp} = G^{sp}(X^{s-2,p}, X^{s-1,p}, X^{s,p}, X^{s+1,p}, X^{s+2,p}) \tag{359}$$

The discretization of (344) is

$$(1-\delta^p_4)X^{sp}_5 + G^{sp} = 0, \text{ for } p \in \mathbb{Z}_5 \tag{360}$$

The state at time t is a root of the function

$$
\begin{aligned}
F : \mathscr{H} &\to \mathscr{H} \\
|X\rangle &\mapsto |F(X)\rangle \\
&= \sum_{s=-2}^{N+1} F^{sp} \left|sp\right\rangle \\
&= \sum_{s=0}^{N-1} \sum_{p\in\mathbb{Z}_5} \left[(1-\delta_4^p) X_5^{sp} + G^{sp}\right] \left|sp\right\rangle \\
&+ |BC\rangle
\end{aligned}
\tag{361}
$$

where $|BC\rangle$ is defined by

$$
\begin{aligned}
|BC\rangle &= \sum_{s=-2}^{-1} \sum_{p\in\mathbb{Z}_5} B^{sp} \left|sp\right\rangle + \sum_{s=N}^{N+1} \sum_{p\in\mathbb{Z}_5} B^{sp} \left|sp\right\rangle \\
BC^{sp} &= \left\langle sp \middle| BC \right\rangle
\end{aligned}
\tag{362}
$$

$$
BC^{-2,p} = \begin{cases}
X^{-2,0} = \Omega_0(-2) & \text{for } p = 0 \\
X^{1,1} - X^{-1,1} = \mathrm{d}s\,\partial_s \Omega_1(0) & \text{for } p = 1 \\
X^{1,2} - X^{-1,2} = \mathrm{d}s\,\partial_s \Omega_2(0) & \text{for } p = 2 \\
X^{-2,3} = f(-2) & \text{for } p = 3 \\
X^{-2,4} = \lambda(-2) & \text{for } p = 4
\end{cases}
\tag{363}
$$

$$BC^{-1,p} = \begin{cases} X^{0,0} = \Omega_0(0) & \text{for } p = 0 \\ X^{0,1} = \Omega_1(0) & \text{for } p = 1 \\ X^{0,2} = \Omega_2(0) & \text{for } p = 2 \\ X^{1,3} - X^{-1,3} = \mathrm{d}s\,\partial_s f(0) & \text{for } p = 3 \\ X^{0,4} = \lambda(0) & \text{for } p = 4 \end{cases} \tag{364}$$

$$BC^{N,p} = \begin{cases} X^{N-1,0} = \Omega_0(N-1) & \text{for } p = 0 \\ X^{N-1,1} = \Omega_1(N-1) & \text{for } p = 1 \\ X^{N-1,2} = \Omega_2(N-1) & \text{for } p = 2 \\ X^{N,3} - X^{N-2,3} = \mathrm{d}s\,\partial_s f(N-1) & \text{for } p = 3 \\ X^{N-1,4} = \lambda(N-1) & \text{for } p = 4 \end{cases} \tag{365}$$

$$BC^{N+1,p} = \begin{cases} X^{N+1,0} = \Omega_0(N+1) & \text{for } p = 0 \\ X^{N,1} - X^{N-2,1} = \mathrm{d}s\,\partial_s \Omega_1(N-1) & \text{for } p = 1 \\ X^{N,2} - X^{N-2,2} = \mathrm{d}s\,\partial_s \Omega_2(N-1) & \text{for } p = 2 \\ X^{N+1,3} = f(N+1) & \text{for } p = 3 \\ X^{N+1,4} = \lambda(N+1) & \text{for } p = 4 \end{cases} \tag{366}$$

Some things you need to calculate Jacobian:

$$\frac{\partial X_5^{sp}}{\partial X^{s'p'}} = \frac{3}{2\,\mathrm{d}t}\delta^s_{s'}\delta^p_{p'} \tag{367}$$

$$\frac{\partial X_q^{sp}}{\partial X^{s'p'}} = \frac{\partial}{\partial X^{s'p'}} \sum_{s''} D_{ss''}^{(q)} X^{s''p} \tag{368}$$

$$= D_{ss'}^{(q)} \delta_{p'}^{p} \tag{369}$$

$$\frac{\partial G^{sp}}{\partial X^{s'p'}} = \sum_{p'',q} \frac{\partial G^p}{\partial X_q^{sp''}} \frac{\partial X_q^{sp''}}{\partial X^{s'p'}} \tag{370}$$

$$= \sum_{q\in\mathbb{Z}_5} \frac{\partial G^p}{\partial X_q^{sp'}} D_{ss'}^{(q)} \tag{371}$$

Note that $D_{ss'}^{(q)} = 0$ and $\frac{\partial G^{sp}}{\partial X^{s'p'}} = 0$ when $|s - s'| > 2$.

$$\frac{\partial}{\partial X^{s'p'}} BC^{-2,p} = \begin{cases} \delta_{s'}^{-2}\delta_{p'}^{0} & \text{for } p = 0 \\ \delta_{s'}^{1}\delta_{p'}^{1} - \delta_{s'}^{-1}\delta_{p'}^{1} & \text{for } p = 1 \\ \delta_{s'}^{1}\delta_{p'}^{2} - \delta_{s'}^{-1}\delta_{p'}^{2} & \text{for } p = 2 \\ \delta_{s'}^{-2}\delta_{p'}^{3} & \text{for } p = 3 \\ \delta_{s'}^{-2}\delta_{p'}^{4} & \text{for } p = 4 \end{cases} \tag{372}$$

$$\frac{\partial}{\partial X^{s'p'}} BC^{-1,p} = \begin{cases} \delta_{s'}^{0}\delta_{p'}^{0} & \text{for } p = 0 \\ \delta_{s'}^{0}\delta_{p'}^{1} & \text{for } p = 1 \\ \delta_{s'}^{0}\delta_{p'}^{2} & \text{for } p = 2 \\ \delta_{s'}^{1}\delta_{p'}^{3} - \delta_{s'}^{-1}\delta_{p'}^{3} & \text{for } p = 3 \\ \delta_{s'}^{0}\delta_{p'}^{4} & \text{for } p = 4 \end{cases} \tag{373}$$

$$\frac{\partial}{\partial X^{s'p'}} BC^{Np} = \begin{cases} \delta^{N-1}_{s'} \delta^{0}_{p'} & \text{for } p = 0 \\ \delta^{N-1}_{s'} \delta^{1}_{p'} & \text{for } p = 1 \\ \delta^{N-1}_{s'} \delta^{2}_{p'} & \text{for } p = 2 \\ \delta^{N}_{s'} \delta^{3}_{p'} - \delta^{N-2}_{s'} \delta^{3}_{p'} & \text{for } p = 3 \\ \delta^{N-1}_{s'} \delta^{4}_{p'} & \text{for } p = 4 \end{cases} \tag{374}$$

$$\frac{\partial}{\partial X^{s'p'}} BC^{N+1,p} = \begin{cases} \delta^{N+1}_{s'} \delta^{0}_{p'} & \text{for } p = 0 \\ \delta^{N}_{s'} \delta^{1}_{p'} - \delta^{N-2}_{s'} \delta^{1}_{p'} & \text{for } p = 1 \\ \delta^{N}_{s'} \delta^{2}_{p'} - \delta^{N-2}_{s'} \delta^{2}_{p'} & \text{for } p = 2 \\ \delta^{N+1}_{s'} \delta^{3}_{p'} & \text{for } p = 3 \\ \delta^{N+1}_{s'} \delta^{4}_{p'} & \text{for } p = 4 \end{cases} \tag{375}$$

$$\frac{\partial BC^{sp}}{\partial X^{s'p'}} = \delta^{p}_{p'} \frac{\partial BC^{sp}}{\partial X^{s'p}} \tag{376}$$

$$\frac{\partial}{\partial X^{s'p'}} \left[(1 - \delta^{p}_{4}) X^{sp}_{5} + G^{sp} \right] = (1 - \delta^{p}_{4}) \frac{3}{2\,\mathrm{d}t} \delta^{s}_{s'} \delta^{p}_{p'} + \sum_{q \in \mathbb{Z}_5} \frac{\partial G^{p}}{\partial X^{sp'}_{q}} D^{(q)}_{ss'} \tag{377}$$

Jacobian:

$$
\begin{aligned}
DF &= \sum_{s=-2}^{N+1} \sum_{s'=-2}^{N+1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle\left\langle s'p'\right| \\
&= \sum_{s\in\mathbb{Z}_N} \sum_{|s'-s|\leq -2} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle\left\langle s'p'\right| \\
&\quad + \sum_{s=-2}^{-1} \sum_{s'=-2}^{1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle\left\langle s'p'\right| \\
&\quad + \sum_{s=N}^{N+1} \sum_{s'=N-2}^{N+1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial F^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle\left\langle s'p'\right|
\end{aligned}
\tag{378}
$$

$$
\begin{aligned}
DF &= \sum_{s\in\mathbb{Z}_N} \sum_{|s'-s|\leq -2} \sum_{p,p'\in\mathbb{Z}_5} \left[(1-\delta_4^p)\frac{3}{2\,\mathrm{d}t}\delta_{s'}^{s}\delta_{p'}^{p} + \sum_{q\in\mathbb{Z}_5} \frac{\partial G^p}{\partial X_q^{sp'}} D_{ss'}^{(q)} \right] \left|sp\right\rangle\left\langle s'p'\right| \\
&\quad + \sum_{s=-2}^{-1} \sum_{s'=-2}^{1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle\left\langle s'p'\right| \\
&\quad + \sum_{s=N}^{N+1} \sum_{s'=N-2}^{N+1} \sum_{p,p'\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p'}} \left|sp\right\rangle\left\langle s'p'\right|
\end{aligned}
\tag{379}
$$

$$
\begin{aligned}
DF &= \sum_{s\in\mathbb{Z}_N} \sum_{p\in\mathbb{Z}_4} \frac{3}{2\,\mathrm{d}t} \left|sp\right\rangle\left\langle sp\right| \\
&\quad + \sum_{s\in\mathbb{Z}_N} \sum_{|s'-s|\leq -2} \sum_{p,p'\in\mathbb{Z}_5} \sum_{q\in\mathbb{Z}_5} \frac{\partial G^p}{\partial X_q^{sp'}} D_{ss'}^{(q)} \left|sp\right\rangle\left\langle s'p'\right| \\
&\quad + \sum_{s=-2}^{-1} \sum_{s'=-2}^{1} \sum_{p\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p}} \left|sp\right\rangle\left\langle s'p\right| \\
&\quad + \sum_{s=N}^{N+1} \sum_{s'=N-2}^{N+1} \sum_{p\in\mathbb{Z}_5} \frac{\partial BC^{sp}}{\partial X^{s'p}} \left|sp\right\rangle\left\langle s'p\right|
\end{aligned}
\tag{380}
$$

$$
\begin{aligned}
DF = &\sum_{s\in\mathbb{Z}_N}\sum_{p\in\mathbb{Z}_4}\frac{3}{2\,\mathrm{d}t}\left|sp\right\rangle\left\langle sp\right| \\
&+\sum_{s\in\mathbb{Z}_N}\sum_{h=-2}^{2}\sum_{p,p'\in\mathbb{Z}_5}\sum_{q\in\mathbb{Z}_5}\frac{\partial G^p}{\partial X_q^{sp'}}D^{(q)}_{s,s+h}\left|sp\right\rangle\left\langle s+h,p'\right| \\
&+\sum_{s=-2}^{-1}\sum_{s'=-2}^{1}\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|sp\right\rangle\left\langle s'p\right| \\
&+\sum_{s=N}^{N+1}\sum_{s'=N-2}^{N+1}\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|sp\right\rangle\left\langle s'p\right|
\end{aligned}
\tag{381}
$$

$$
\boxed{
\begin{aligned}
DF = &\sum_{s\in\mathbb{Z}_N}\left|s\right\rangle\left\langle s\right|\otimes\left(\sum_{p\in\mathbb{Z}_4}\frac{3}{2\,\mathrm{d}t}\left|p\right\rangle\left\langle p\right|\right) \\
&+\sum_{s\in\mathbb{Z}_N}\sum_{h=-2}^{2}\left|s\right\rangle\left\langle s+h\right|\otimes\left(\sum_{p,p'\in\mathbb{Z}_5}\sum_{q\in\mathbb{Z}_5}\frac{\partial G^p}{\partial X_q^{sp'}}D^{(q)}_{s,s+h}\left|p\right\rangle\left\langle p'\right|\right) \\
&+\sum_{s=-2}^{-1}\sum_{s'=-2}^{1}\left|s\right\rangle\left\langle s'\right|\otimes\left(\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|p\right\rangle\left\langle p\right|\right) \\
&+\sum_{s=N}^{N+1}\sum_{s'=N-2}^{N+1}\left|s\right\rangle\left\langle s'\right|\otimes\left(\sum_{p\in\mathbb{Z}_5}\frac{\partial BC^{sp}}{\partial X^{s'p}}\left|p\right\rangle\left\langle p\right|\right)
\end{aligned}
}
\tag{382}
$$

$$
\boxed{
\begin{aligned}
&\left|Y_0\right\rangle = 2\left|X^{t-1}\right\rangle - \left|X^{t-2}\right\rangle \\
&\left|F(Y_\alpha)\right\rangle + DF(Y_\alpha)(\left|Y_{\alpha+1}\right\rangle - \left|Y_\alpha\right\rangle)
\end{aligned}
}
\tag{383}
$$

6.1.1subsubsectionsection=0 . Nonzero p, p' blocks

Let $s \in \mathbb{Z}_N$

$$
\sum_{s\in\mathbb{Z}_N}\sum_{h=-2}^{2}\left|s\right\rangle\left\langle s+h\right|\otimes\left(\sum_{p,p'\in\mathbb{Z}_5}\sum_{q\in\mathbb{Z}_5}\frac{\partial G^p}{\partial X_q^{sp'}}D^{(q)}_{s,s+h}\left|p\right\rangle\left\langle p'\right|\right) \tag{384}
$$

$$
= \sum_{p,p'\in\mathbb{Z}_5}\sum_{h=-2}^{2}\sum_{q\in\mathbb{Z}_5}D^{(q)}_{0,h}\left[\sum_{s\in\mathbb{Z}_N}\left|s\right\rangle\left\langle s+h\right|\frac{\partial G^p}{\partial X_q^{sp'}}\right]\otimes\left|p\right\rangle\left\langle p'\right| \tag{385}
$$

so, the diagonal of block (p, p') with offset h is given by

$$\sum_{q\in\mathbb{Z}_5} D_{0,h}^{(q)} \begin{bmatrix} 0 \\ 0 \\ \frac{\partial G^p}{\partial X_q^{0,p'}} \\ \vdots \\ \frac{\partial G^p}{\partial X_q^{sp'}} \\ \vdots \\ \frac{\partial G^p}{\partial X_q^{N,p'}} \\ 0 \\ 0 \end{bmatrix} \tag{386}$$

6.1.2subsubsectionsection=0 . Frame and centerline (old notation)

$$\boldsymbol{v}^{[0,j]} = \frac{1}{\alpha_1}\partial_s\lambda^{[0,j]}\boldsymbol{e}_1 - \frac{B_2}{\alpha_2}\partial_s^2\omega_3^{[0,j]}\boldsymbol{e}_2 + \frac{B_2}{\alpha_2}\partial_s^2\omega_2^{[0,j]}\boldsymbol{e}_3 \tag{387}$$

$$\boldsymbol{\Omega}^{[0,j]} = \frac{B_1}{\beta}\partial_s\omega_1^{[0,j]}\boldsymbol{e}_1 - \frac{B_2}{\alpha_2}\partial_s^3\omega_2^{[0,j]}\boldsymbol{e}_2 - \frac{B_2}{\alpha_2}\partial_s^3\omega_3^{[0,j]}\boldsymbol{e}_3 \tag{388}$$

$$\psi_t^{[0,j]} = \begin{pmatrix} \hat{\boldsymbol{\Omega}}^{[0,j]} & \boldsymbol{v}^{[0,j]} \\ 0 & 0 \end{pmatrix} \tag{389}$$

$$\psi_s^{[j,k+1]} = \begin{pmatrix} \hat{\boldsymbol{\omega}}^{[j,k+1]} & \boldsymbol{e}_1 \\ 0 & 0 \end{pmatrix} \tag{390}$$

- timestep $t \to t + \mathrm{d}t$

$$\rho^{[0,k+1]} = \rho^{[0,k]} \exp\left(\frac{\mathrm{d}t}{2}\left[\psi_t^{[0,k+1]} + \psi_t^{[0,k]}\right]\right) \tag{391}$$

$$\rho^{[j+1,k+1]} = \rho^{[j,k+1]} \exp\left(\frac{\mathrm{d}s}{2}\left[\psi_s^{[j+1,k+1]} + \psi_s^{[j,k+1]}\right]\right) \tag{392}$$

Required data for timestep $t \to t + \mathrm{d}t$:

- $s = 0$, time t: $Q, x, \partial_s \omega_1, \partial_s^2 \boldsymbol{\omega}_\perp, \partial_s^3 \boldsymbol{\omega}_\perp, \partial_s \lambda$
- $s = 0$, time $t + \mathrm{d}t$: $\partial_s \omega_1, \partial_s^2 \boldsymbol{\omega}_\perp, \partial_s^3 \boldsymbol{\omega}_\perp, \partial_s \lambda$
- all s, time $t + \mathrm{d}t$: $\omega_1, \boldsymbol{\omega}_\perp$
- Parameters: $B_1, B_2, \alpha_1, \alpha_2, \beta$

Asectionsection=0 . Ordered exponentials

Suppose G is a finite dimensional Lie group and let $\mathscr{A}$ be the G-bundle over a compact domain $S \subset \mathbb{R}^p$.

Theorem 1. *Let g, h be sections of $\mathscr{A}$, which satisfy $g(s^1, \ldots, s^p) = g(0, \ldots, 0, s^k, 0 \ldots, 0)\, h(s^1, \ldots, s^p)$ and let $\psi_g = \psi_{g,n}\, \mathrm{d}s^n$, $\psi_h = \psi_{h,n}\, \mathrm{d}s^n$ be the Maurer-Cartan forms for g, h. Then*

$$h(0, \ldots, 0, s^k, 0 \ldots, 0) = 1 \tag{393}$$

$$\psi_{g,j} = \begin{cases} \psi_{h,j} & \textit{for } j \neq k \\ \psi_{h,k} + h^{-1}\psi_{g,k}(\ldots,0,s^k,0\ldots)h & \textit{for } j = k \end{cases} \tag{394}$$

$$\psi_{h,k}(0,\ldots,0,s^k,0\ldots,0) = 0 \tag{395}$$

$$g(s) = g(0)\,Oexp\left[\psi_{g,k}(0,\ldots,s^k,\ldots,0);s^k\right]h(s) \tag{396}$$

Proof: Equation (393) follows from evaluating the defining eqn of h at $s = (0,\ldots,0,s^k,0\ldots,0)$. Equation (394) follows from taking derivatives, and Equation (395) follows from evaluating Equation (394) at $s = (0,\ldots,0,s^k,0\ldots,0)$ and using Equation (393). □

Theorem 2. *Let g be a section of $\mathscr{A}$ with Maurer-Cartan form $\psi = \psi_n\,\mathrm{d}s^n$. Then,*

$$g(s) = g(0)\,Oexp\left[\psi_1(\cdot,0\ldots);s^1\right]Oexp\left[\psi_2(s^1,\cdot,0\ldots);s^2\right]\cdots Oexp\left[\psi_p(s^1,\ldots,s^{p-1},\cdot);s^p\right] \tag{397}$$

$$= g(0)\prod_{k=1}^{p} Oexp\left[\psi_1(s^1,\cdots,s^{k-1},\cdot,0\ldots);s^k\right] \tag{398}$$

Proof: By induction... □

Ex: Suppose g represents the state of a Kirchoff rod, so, $S \subset \mathbb{R}^{1+1}$, $G = SE(3)$. Given the Maurer Cartan form and the initial configuration of a single cross-section, we can reconstruct the configuration of a Kirchoff rod at later times by solving a sequence of initial value problems. Define $h(s,t)$ to be the transformation which maps cross-section 0 onto cross-section s at time t.

$$g(s,t) = g(0,t)\,h(s,t) \tag{399}$$

$$h(0,t) = 1 \tag{400}$$

$$\mathrm{d}g =: g\psi_g \tag{401}$$

$$\mathrm{d}h =: h\psi_h \tag{402}$$

$$\psi_{g,s}(s,t) = \psi_{h,s}(s,t) \tag{403}$$

$$\psi_{g,t}(s,t) = \psi_{h,t}(s,t) + h^{-1}(s,t)\psi_{g,t}(0,t)h(s,t) \tag{404}$$

$$\psi_{h,t}(0,t) = 0 \tag{405}$$

$$h(s,t) = \mathrm{Oexp}\left[\psi_{g,s}(\cdot,t); s\right] \tag{406}$$

$$g(0,t) = g(0,0)\mathrm{Oexp}\left[\psi_{g,t}(0,\cdot); t\right] \tag{407}$$

$$g(s,t) = g(0,0)\mathrm{Oexp}\left[\psi_{g,t}(0,\cdot); t\right]\mathrm{Oexp}\left[\psi_{g,s}(\cdot,t); s\right] \tag{408}$$

$$\psi_s = \begin{pmatrix} \hat{\omega} & e_1 \\ 0 & 0 \end{pmatrix} \tag{409}$$

$$\psi_t = \begin{pmatrix} \hat{\Omega} & \nu \\ 0 & 0 \end{pmatrix} \tag{410}$$

A.1subsectionsection=0 . References

- [3]

Bsectionsection=0 . Finite difference formulas

$$\boxed{\frac{3\omega^{[k+1]} - 4\omega^{[k]} + \omega^{[k-1]}}{2\,\mathrm{d}t} = \frac{\partial\omega}{\partial t}^{[k+1]} - \frac{1}{3}\,\mathrm{d}t^2\,\frac{\partial^3\omega}{\partial t^3}^{[k+1]} + O(\mathrm{d}t^3)} \tag{411}$$

Proof:

$$\begin{aligned} -4\omega(t+h-h) &= -4\left(\omega(t+h) - h\dot{\omega}(t+h) + \frac{1}{2}h^2\ddot{\omega}(t+h) - \frac{1}{6}h^3\dddot{\omega}(t+h)\right) + O(h^4) \\ &= -4\omega(t+h) + 4h\dot{\omega}(t+h) - 2h^2\ddot{\omega}(t+h) + \frac{2}{3}h^3\dddot{\omega}(t+h) + O(h^4) \end{aligned}$$

$$\omega(t+h-2h) = \omega(t+h) - 2h\dot{\omega}(t+h) + 2h^2\ddot{\omega}(t+h) - \frac{4}{3}h^3\dddot{\omega}(t+h) + O(h^4)$$

$$-4\omega(t+h-h) + \omega(t+h-2h) = -3\omega(t+h) + 2h\dot{\omega}(t+h) + 2h^2\ddot{\omega}(t+h) - \frac{2}{3}h^3\dddot{\omega}(t+h) + O(h^4)$$

$$\frac{3\omega(t+h) - 4\omega(t+h-h) + \omega(t+h-2h)}{2h} = \dot{\omega}(t+h) - \frac{1}{3}h^2\dddot{\omega}(t+h) + O(h^3)$$

□

$$\boxed{\frac{\omega^{[j+1,k+1]} - 2\omega^{[j,k+1]} + \omega^{[j-1,k+1]}}{\mathrm{d}s^2} = \frac{\partial^2 \omega}{\partial s^2}^{[j,k+1]} + \frac{1}{12}\,\mathrm{d}s^2\,\omega^{[j,k+1]''''} + O(\mathrm{d}s^3)} \tag{412}$$

Proof:

$$\omega(s+h) = \omega(s) + h\omega'(s) + \frac{1}{2}h^2\omega''(s) + \frac{1}{6}h^3\omega'''(s) + \frac{1}{24}h^4\omega''''(s) + O(h^5)$$

$$\omega(s-h) = \omega(s) - h\omega'(s) + \frac{1}{2}h^2\omega''(s) - \frac{1}{6}h^3\omega'''(s) + \frac{1}{24}h^4\omega''''(s) + O(h^5)$$

$$\omega(s+h) + \omega(s-h) = 2\omega(s) + h^2\omega''(s) + \frac{1}{12}h^4\omega''''(s) + O(h^5)$$

$$\frac{\omega(s+h) - 2\omega(s) + \omega(s-h)}{h^2} = \omega''(s) + \frac{1}{12}h^2\omega''''(s) + O(h^3)$$

□

$$\boxed{\omega^{[k+1]} = 2\omega^{[k]} - \omega^{[k-1]} + \mathrm{d}t^2\,\ddot{\omega}(t) + O(\mathrm{d}t^4)} \tag{413}$$

Proof:

$$\frac{\omega(t+h) - 2\omega(t) + \omega(t-h)}{h^2} = \ddot{\omega}(t) + O(h^2)$$

$$2\omega(t) - \omega(t-h) = \omega(t+h) - h^2\ddot{\omega}(t) + O(h^4)$$

□

$$\boxed{\rho(s)\exp\left(\frac{h}{2}\left[\psi(s)+\psi(s+h)\right]\right) = \rho(s+h) + \frac{h^3}{12}\rho(s)\left(\psi'' - \psi\psi' + \psi'\psi\right) + O(h^4)} \tag{414}$$

Proof:

$$\begin{aligned}
\rho' &= \rho\psi \\
\rho'' &= \rho\left(\psi' + \psi^2\right) \\
\rho''' &= \rho\left(\psi'' + \psi'\psi + 2\psi\psi' + \psi^3\right)
\end{aligned}$$

$$\begin{aligned}
\rho(s+h) &= \rho(s) + h\rho'(s) + \frac{1}{2}h^2\rho''(s) + \frac{1}{6}h^3\rho'''(s) + O(h^4) \\
&= \rho\left[1 + h\psi + \frac{1}{2}h^2\left(\psi' + \psi^2\right) + \frac{1}{6}h^3\left(\psi'' + \psi'\psi + 2\psi\psi' + \psi^3\right)\right] + O(h^4)
\end{aligned}$$

$$\begin{aligned}
\exp\left(\frac{h}{2}\left[\psi(s)+\psi(s+h)\right]\right) &= \exp\left(\frac{h}{2}\left[2\psi(s) + h\psi'(s) + \frac{1}{2}h^2\psi''(s) + O(h^3)\right]\right) \\
&= \exp\left(h\psi + \frac{1}{2}h^2\psi' + \frac{1}{4}h^3\psi'' + O(h^4)\right) \\
&= 1 + h\psi + \frac{1}{2}h^2\psi' + \frac{1}{4}h^3\psi'' + \frac{1}{2}\left[h\psi + \frac{1}{2}h^2\psi'\right]^2 + \frac{1}{6}\left[h\psi\right]^3 + O(h^4) \\
&= 1 + h\psi + \frac{1}{2}h^2\psi' + \frac{1}{4}h^3\psi'' + \frac{1}{2}\left[h^2\psi^2 + \frac{1}{2}h^3\psi\psi' + \frac{1}{2}h^3\psi'\psi\right] + \frac{1}{6}h^3\psi^3 + O(h^4) \\
&= 1 + h\psi + \frac{1}{2}h^2\left[\psi' + \psi^2\right] + h^3\left[\frac{1}{4}\psi'' + \frac{1}{4}\psi\psi' + \frac{1}{4}\psi'\psi + \frac{1}{6}\psi^3\right] + O(h^4)
\end{aligned}$$

$$\begin{aligned}
\rho(s)\exp\left(\frac{h}{2}\left[\psi(s)+\psi(s+h)\right]\right) &= \rho(s)\left(1+h\psi+\frac{1}{2}h^2\left[\psi'+\psi^2\right]+h^3\left[\frac{1}{4}\psi''+\frac{1}{4}\psi\psi'+\frac{1}{4}\psi'\psi+\frac{1}{6}\psi^3\right]\right)+O(h^4)\\
&= \rho(s+h)+h^3\rho(s)\left(\frac{1}{4}\psi''+\frac{1}{4}\psi\psi'+\frac{1}{4}\psi'\psi+\frac{1}{6}\psi^3\right)\\
&\quad -h^3\rho(s)\left(\frac{1}{6}\psi''+\frac{1}{6}\psi'\psi+\frac{1}{3}\psi\psi'+\frac{1}{6}\psi^3\right)+O(h^4)\\
&= \rho(s+h)+\frac{h^3}{12}\rho(s)\left(\psi''-\psi\psi'+\psi'\psi\right)+O(h^4)
\end{aligned}$$

□

Csectionsection=0 . Biharmonic stuff

C.1subsectionsection=0 . Green function

$$\partial_s^4 G(s,s') = \delta(s-s') \tag{415}$$

$$G(0,s') = G(1,s') = 0 \tag{416}$$

$$\partial_s G(s,s')\big|_{s=0} = \partial_s G(s,s')\big|_{s=1} = 0 \tag{417}$$

$$G(x,y) = \begin{cases} \frac{1}{12}(1-x)(1-y)^2x^2y+\frac{1}{24}(y-x)(1-y)^2x^2, & \text{for } x<y \\ \frac{1}{12}(1-y)(1-x)^2y^2x+\frac{1}{24}(x-y)(1-x)^2y^2, & \text{for } y<x \end{cases} \tag{418}$$

$$p(x,y) = \frac{1}{12}(1-x)(1-y)^2x^2y+\frac{1}{24}(y-x)(1-y)^2x^2 \tag{419}$$

$$G(x,y) = p(x,y)\Theta(y-x)+p(y,x)\Theta(x-y) \tag{420}$$

C.2subsectionsection=0 . Free-free eigenfunctions

$$\partial_s^4 \psi_k = k^4 \psi_k \tag{421}$$

$$\psi_k = a_1 \cos(ks) + a_2 \sin(ks) + a_3 \cosh(ks) + a_4 \sinh(ks) \tag{422}$$

$$\partial_s \psi_k = -k a_1 \sin(ks) + k a_2 \cos(ks) + k a_3 \sinh(ks) + k a_4 \cosh(ks) \tag{423}$$

$$\psi_k(0) = \psi_k(1) = 0, \qquad \partial_s \psi_k(0) = \partial_s \psi_k(1) = 0 \tag{424}$$

$$\psi_k = [\sinh(k) - \sin(k)]\,[\cos(ks) - \cosh(ks)] + [\cos(k) - \cosh(k)]\,[\sin(ks) - \sinh(ks)] \tag{425}$$

$$\cos(k)\cosh(k) = 1 \tag{426}$$

$$k_0 \approx 4.73004074 \tag{427}$$

$$k_1 \approx 7.85320462 \tag{428}$$

$$k_2 \approx 10.99560784 \tag{429}$$

$$k_3 \approx 14.13716549 \tag{430}$$

$$k_4 \approx 17.27875966 \tag{431}$$

$$k_5 \approx 20.42035225 \tag{432}$$

$$k_6 \approx 23.5619449 \tag{433}$$

$$k_7 \approx 26.70353756 \tag{434}$$

$$k_8 \approx 29.84513021 \tag{435}$$

$$k_9 \approx 32.98672286 \tag{436}$$

$$k_{10} \approx 36.12831552 \tag{437}$$

$$k_{11} \approx 39.26990817 \tag{438}$$

$$k_{12} \approx 42.41150082 \tag{439}$$

$$k_{13} \approx 45.55309348 \tag{440}$$

$$k_{14} \approx 48.69468613 \tag{441}$$

$$k_{15} \approx 51.83627878 \tag{442}$$

$$k_{16} \approx 54.97787144 \tag{443}$$

$$k_{17} \approx 58.11946409 \tag{444}$$

$$k_{18} \approx 61.26105675 \tag{445}$$

$$k_{19} \approx 64.4026494 \tag{446}$$

$$k_0/\pi \approx 1.50561873 \tag{447}$$

$$k_1/\pi \approx 2.49975267 \tag{448}$$

$$k_2/\pi \approx 3.50001068 \tag{449}$$

$$k_3/\pi \approx 4.49999954 \tag{450}$$

$$k_4/\pi \approx 5.50000002 \tag{451}$$

$$k_5/\pi \approx 6.5 \tag{452}$$

$$k_6/\pi \approx 7.5 \tag{453}$$

$$k_7/\pi \approx 8.5 \tag{454}$$

$$k_8/\pi \approx 9.5 \tag{455}$$

$$k_9/\pi \approx 10.5 \tag{456}$$

$$k_{10}/\pi \approx 11.5 \tag{457}$$

$$k_{11}/\pi \approx 12.5 \tag{458}$$

$$k_{12}/\pi \approx 13.5 \tag{459}$$

$$k_{13}/\pi \approx 14.5 \tag{460}$$

$$k_{14}/\pi \approx 15.5 \tag{461}$$

$$k_{15}/\pi \approx 16.5 \tag{462}$$

$$k_{16}/\pi \approx 17.5 \tag{463}$$

$$k_{17}/\pi \approx 18.5 \tag{464}$$

$$k_{18}/\pi \approx 19.5 \tag{465}$$

$$k_{19}/\pi \approx 20.5 \tag{466}$$

Proof:

$$\psi_k(0) = a_1 + a_3 = 0, \qquad \partial_s \psi_k(0) = k a_2 + k a_4 = 0 \tag{467}$$

$$\psi_k = a_1 \left[\cos(ks) - \cosh(ks)\right] + a_2 \left[\sin(ks) - \sinh(ks)\right] \tag{468}$$

$$\partial_s \psi_k = -k a_1 \left[\sin(ks) + \sinh(ks)\right] + k a_2 \left[\cos(ks) - \cosh(ks)\right] \tag{469}$$

$$\psi_k(1) = a_1 \left[\cos(k) - \cosh(k)\right] + a_2 \left[\sin(k) - \sinh(k)\right] = 0 \tag{470}$$

$$\psi_k = -\left[\sin(k) - \sinh(k)\right]\left[\cos(ks) - \cosh(ks)\right] + \left[\cos(k) - \cosh(k)\right]\left[\sin(ks) - \sinh(ks)\right] \tag{471}$$

$$\partial\psi_k = k\left[\sin(k) - \sinh(k)\right]\left[\sin(ks) + \sinh(ks)\right] + k\left[\cos(k) - \cosh(k)\right]\left[\cos(ks) - \cosh(ks)\right] \tag{472}$$

$$\partial\psi_k(1) = k\left[\sin(k) - \sinh(k)\right]\left[\sin(k) + \sinh(k)\right] + k\left[\cos(k) - \cosh(k)\right]\left[\cos(k) - \cosh(k)\right] \tag{473}$$

$$= 2k\left[1 - \cos(k)\cosh(k)\right] \tag{474}$$

$$\cos(k)\cosh(k) = 1 \tag{475}$$

□

C.2.1subsubsectionsection=0 . Free-free alt rep

$$\Psi_+ = \cosh(k/2)\cos(k[s-1/2]) - \cos(k/2)\cosh(k[s-1/2]) \tag{476}$$

$$\sin(k/2) + \cos(k/2)\tanh(k/2) = 0 \tag{477}$$

$$k_0, k_2, k_4, \ldots \tag{478}$$

$$\langle \Psi_+ | \Psi_+ \rangle = \frac{1}{2}\left[\cosh^2(k/2) + \cos^2(k/2)\right] \tag{479}$$

Normalized

$$\psi_+ = \frac{\sqrt{2}}{\sqrt{1+\frac{\cos^2(k/2)}{\cosh^2(k/2)}}}\cos(k(s-1/2)) - \frac{\sqrt{2}\cos(k/2)}{(1+e^{-k})\sqrt{1+\frac{\cos^2(k/2)}{\cosh^2(k/2)}}}\left(e^{k(s-1)} + e^{-ks}\right) \tag{480}$$

$$\Psi_- = -\sinh(k/2)\sin(k[s-1/2]) + \sin(k/2)\sinh(k[s-1/2]) \tag{481}$$

$$-\sin(k/2) + \cos(k/2)\tanh(k/2) = 0 \tag{482}$$

$$k_1, k_3, k_5, \ldots \tag{483}$$

$$\langle \Psi_- | \Psi_- \rangle = \frac{1}{2}\left[\sinh^2(k/2) - \sin^2(k/2)\right] \tag{484}$$

Normalized

$$\psi_- = \frac{-\sqrt{2}}{\sqrt{1-\frac{\sin^2(k/2)}{\sinh^2(k/2)}}}\sin(k(s-1/2)) + \frac{\sqrt{2}\sin(k/2)}{(1-e^{-k})\sqrt{1-\frac{\sin^2(k/2)}{\sinh^2(k/2)}}}\left(e^{k(s-1)} - e^{-ks}\right) \tag{485}$$

C.2.2subsubsectionsection=0 . Asymptotics

$$\frac{\cos(ks)-\cosh(ks)}{\cos(k)-\cosh(k)} + \frac{\sin(ks)-\sinh(ks)}{\sinh(k)-\sin(k)} \tag{486}$$

$$\cos(k) \sim 0 \tag{487}$$

$$k_n = (j+3/2)\pi + \varepsilon_j \tag{488}$$

Dsectionsection=0 . Green function and some spectral stuff

Let $\hat{H}$ be a local self adjoint operator.

$$\langle x|\hat{H}\left|x'\right\rangle = H(x)\delta(x-x') \tag{489}$$

$$\hat{H}^\dagger = \hat{H} \tag{490}$$

Let $|k\rangle$ represent the eigenvector of $\hat{H}$ with eigenvalue k.

$$\hat{H}|k\rangle = k|k\rangle \tag{491}$$

The right inverse of $\hat{H}$ is called a Green function

$$\hat{H}\hat{G} = 1 \tag{492}$$

$$\langle x| \hat{G} \left| x' \right\rangle = G(x, x') \tag{493}$$

$$\langle x| \hat{H}\hat{G} \left| x' \right\rangle = H(x)G(x, x')\delta(x - x') \tag{494}$$

$$\hat{G} = \sum_k \frac{1}{k} |k\rangle \langle k| \tag{495}$$

Esectionsection=0 . Laplace/Poisson equation stuff

E.0.1subsubsectionsection=0 . Dirichlet eigenfunctions

Let $0 < s < 1$ and $n = 1, 2, 3, \ldots$

$$\left\langle s \middle| \phi_n \right\rangle = \phi_n(s) = \sqrt{2}\sin(n\pi s) \tag{496}$$

$$\left\langle \phi_m \middle| \phi_n \right\rangle = \int_0^1 \phi_m(s)\phi_n(s)\,\mathrm{d}s = \delta_{mn} \tag{497}$$

$$\partial_s^2 \phi_n = -(n\pi)^2 \phi_n \tag{498}$$

The eigenfunctions can also be separated into even/odd functions about $s = 1/2$ representation. Let $m = 1, 2, 3, \ldots$

$$\left| \phi_{2m} \right\rangle := (-1)^m \left| \phi_m^- \right\rangle \tag{499}$$

$$\left| \phi_{2m-1} \right\rangle := (-1)^{m-1} \left| \phi_m^+ \right\rangle \tag{500}$$

$$\phi_m^+(s) = \sqrt{2}\cos((2m-1)\pi(s-1/2)) \tag{501}$$

$$\phi_m^-(s) = \sqrt{2}\cos(2m\pi(s-1/2)) \tag{502}$$

Note the parity of the index is opposite the parity of the functions!

Proof:

$$\sin(n\pi s) = \sin(n\pi(s-1/2) + n\pi/2) \tag{503}$$

$$= \cos(n\pi/2)\sin(n\pi(s-1/2)) + \sin(n\pi/2)\cos(n\pi(s-1/2)) \tag{504}$$

$$= \begin{cases} \cos(n\pi/2)\sin(n\pi(s-1/2)), & n \text{ even} \\ \sin(n\pi/2)\cos(n\pi(s-1/2)), & n \text{ odd} \end{cases} \tag{505}$$

For $n = 2m, m = n/2$ we have

$$\cos(n\pi/2) = \cos(m\pi) = (-1)^m = (-1)^{n/2} \tag{506}$$

$$\sin(n\pi s) = (-1)^{n/2}\sin(n\pi(s-1/2)). \tag{507}$$

and for $n = 2m-1, m = (n+1)/2$ we have

$$\sin(n\pi/2) = \sin((2m-1)\pi/2) = (-1)^{m-1} = (-1)^{(n-1)/2} \tag{508}$$

$$\sin(n\pi s) = (-1)^{(n-1)/2}\cos(n\pi(s-1/2)). \tag{509}$$

$$\sin(n\pi s) = \begin{cases} (-1)^{n/2}\sin(n\pi(s-1/2)), & n \text{ even} \\ (-1)^{(n-1)/2}\cos(n\pi(s-1/2)), & n \text{ odd} \end{cases} \tag{510}$$

□

E.0.2subsubsectionsection=0 . Dirichlet Green function

Let $s, s' \in (0,1)$.

$$\partial_s^2 G_{s'} = \delta_{s'} \tag{511}$$

$$G_{s'}(0) = G_{s'}(1) = 0 \tag{512}$$

$$G_{s'}(s) = \begin{cases} (s'-1)s, & \text{for } s < s' \\ s'(s-1), & \text{for } s > s' \end{cases} \tag{513}$$

$$= ss' - s\Theta(s'-s) - s'\Theta(s-s') \tag{514}$$

Proof:

$$G_z(s) = \begin{cases} a_- s + b_-, & \text{for } s < z \\ a_+ s + b_+, & \text{for } s > z \end{cases} \tag{515}$$

$$G_z(0) = 0 \Rightarrow b_- = 0 \tag{516}$$

$$G_z(1) = 0 \Rightarrow b_+ = -a_+ \tag{517}$$

$$G_z(z^+) = G_z(z^-) \Rightarrow a_+ = z, a_- = (z-1) \tag{518}$$

□

Fourier space representation

$$\hat{G} = \sum_{n=1}^{\infty} -\frac{1}{n^2\pi^2} \left|\phi_n\right\rangle \left\langle\phi_n\right| \tag{519}$$

$$G(s,s') = \langle s|\,\hat{G}\,|s'\rangle = \sum_{n=1}^{\infty} -\frac{2}{n^2\pi^2} \sin(n\pi s)\sin\left(n\pi s'\right) \tag{520}$$

Fsectionsection=0 . Some Fourier stuff

$$\sin\left(x+y\right) = \sin(x)\cos\left(y\right) + \sin\left(y\right)\cos(x) \tag{521}$$

$$\cos\left(x+y\right) = \cos(x)\cos\left(y\right) - \sin\left(y\right)\sin(x) \tag{522}$$

$$\left\langle\psi\middle|\phi\right\rangle = \int_0^1 \psi^*(x)\phi(x)\,\mathrm{d}x \tag{523}$$

$$\theta_p(x) = \cos\left(p(x-1/2)\right) \tag{524}$$

$$\phi_p(x) = \sin\left(p(x-1/2)\right) \tag{525}$$

$$\mu_p(x) = \cosh\left(p(x-1/2)\right) \tag{526}$$

$$\nu_p(x) = \sinh\left(p(x-1/2)\right) \tag{527}$$

$$\left\langle\theta_p\middle|\theta_q\right\rangle = 2\frac{p\sin\left(p/2\right)\cos\left(q/2\right) - q\cos\left(p/2\right)\sin\left(q/2\right)}{p^2-q^2} \tag{528}$$

$$\langle \theta_p | \mu_q \rangle = 2 \frac{p \sin(p/2) \cosh(q/2) + q \cos(p/2) \sinh(q/2)}{p^2 + q^2} \tag{529}$$

$$\langle \phi_p | \phi_q \rangle = 2 \frac{q \sin(p/2) \cos(q/2) - p \cos(p/2) \sin(q/2)}{p^2 - q^2} \tag{530}$$

$$\langle \phi_p | \nu_q \rangle = 2 \frac{q \sin(p/2) \cosh(q/2) - p \cos(p/2) \sinh(q/2)}{p^2 + q^2} \tag{531}$$

$$\langle \theta_p | \phi_q \rangle = \langle \theta_p | \nu_q \rangle = \langle \phi_p | \mu_q \rangle = \langle \mu_p | \nu_q \rangle = 0 \tag{532}$$

$$\langle \phi_p | \phi_q \rangle = 2 \frac{q \sin(p/2) \cos(q/2) - p \cos(p/2) \sin(q/2)}{p^2 - q^2} \tag{533}$$

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